

Table 22. Reheat temperature rise — 25 tons (continued)

Airflow (cfm)	Ent DB (°F)	Entering Wet Bulb (°F)											
		51				55				59			
		Lvg Evap DB (°F)		Lvg Reheat DB (°F)		Lvg Evap DB (°F)		Lvg Reheat DB (°F)		Lvg Evap DB (°F)		Lvg Reheat DB (°F)	
		MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX
11000	60	44.3	42.9	48.1	58.3	47.5	45.8	51.7	62.7	50.9	49.2	55.7	67.6
	65	47.5	45.7	51.2	60.3	48.6	47.1	52.7	63.6	51.9	50.0	56.5	68.2
	70	51.4	49.4	55.4	64.9	51.5	49.2	55.5	65.0	53.0	51.3	57.5	69.0
	75	55.4	53.1	59.7	69.4	55.5	53.2	59.7	69.5	55.6	53.3	59.8	70.2
12000	60	45.4	43.8	48.8	58.0	48.2	46.7	52.2	62.3	51.6	50.0	56.1	67.1
	65	48.7	47.0	52.1	60.6	49.7	48.0	53.4	63.2	52.7	50.9	57.0	67.8
	70	52.7	50.7	56.4	65.2	52.8	50.8	56.5	65.3	54.0	52.2	58.1	68.6
	75	56.8	54.5	60.8	69.8	56.8	54.5	60.8	69.8	56.9	54.4	60.9	69.9
Ambient DB 70°F													
5000	60	35.0	32.7 ^(a)	42.3	62.0	38.5	36.0	46.5	67.5	42.3	39.6	50.9	73.1
	65	35.6	33.4 ^(a)	42.9	62.5	39.0	36.5	46.9	67.9	42.8	40.1	51.4	73.5
	70	37.0	34.6 ^(a)	44.1	63.4	39.7	37.2	47.5	68.4	43.2	40.6	51.8	73.9
	75	40.5	37.5	47.9	66.8	40.8	37.9	48.5	68.8	43.9	41.3	52.4	74.3
6000	60	37.6	35.5	44.0	60.9	41.1	38.7	48.1	66.2	44.9	42.3	52.5	71.8
	65	38.4	36.3	44.7	61.5	41.8	39.3	48.7	66.7	45.5	42.9	53.1	72.3
	70	41.2	38.0	47.4	62.8	42.6	40.2	49.4	67.3	46.1	43.5	53.6	72.7
	75	44.6	41.6	51.2	67.4	44.4	41.7	51.0	68.4	46.8	44.3	54.2	73.2
7000	60	39.7	37.6	45.3	60.2	43.1	40.8	49.3	65.3	46.9	44.3	53.7	70.6
	65	41.0	38.5	46.5	60.7	43.9	41.6	50.0	65.9	47.5	45.0	54.3	71.2
	70	44.3	41.7	49.9	63.5	45.1	42.6	51.1	66.4	48.3	45.8	55.0	71.8
	75	47.8	44.9	53.8	68.0	47.9	44.5	53.9	67.8	49.0	46.7	55.6	72.3
8000	60	41.3	39.4	46.4	59.6	44.6	42.5	50.3	64.5	48.6	45.9	54.8	69.6
	65	42.9	41.0	47.8	60.6	45.6	43.4	51.1	65.2	49.1	46.7	55.2	70.3
	70	46.7	44.2	51.8	64.0	47.1	44.9	52.4	66.1	50.0	47.6	56.0	71.0
	75	50.5	47.6	55.9	68.5	50.5	47.7	56.0	68.6	51.3	48.6	57.1	71.5
9000	60	42.6	40.8	47.2	59.1	45.9	43.9	51.0	63.8	49.7	47.2	55.4	68.8
	65	45.0	42.7	49.3	60.3	46.9	44.9	51.9	64.6	50.3	48.0	55.9	69.5
	70	48.8	46.3	53.5	64.4	48.6	46.7	53.4	65.6	51.3	49.1	56.8	70.2
	75	52.6	49.8	57.7	68.9	52.7	49.9	57.8	69.0	52.9	50.7	58.2	71.2
10000	60	43.7	42.1	47.9	58.7	46.9	45.0	51.6	63.2	50.6	48.5	55.8	68.4
	65	46.6	44.1	50.6	60.0	48.0	46.2	52.6	64.1	51.4	49.2	56.5	68.8
	70	50.5	48.1	54.8	64.7	50.6	48.1	54.9	65.3	52.4	50.3	57.4	69.6
	75	54.5	51.7	59.2	69.3	54.6	51.8	59.2	69.4	54.4	52.2	59.2	70.8
11000	60	44.6	43.1	48.5	58.3	47.8	46.0	52.2	62.8	51.4	49.4	56.3	67.7
	65	47.9	45.9	51.7	60.4	49.1	47.2	53.3	63.6	52.3	50.2	57.1	68.3
	70	52.0	49.6	56.0	65.0	52.0	49.4	56.1	65.0	53.5	51.4	58.1	69.1
	75	56.1	53.3	60.4	69.7	56.1	53.4	60.5	69.7	56.2	53.5	60.6	70.4
12000	60	45.7	44.0	49.2	58.0	48.6	46.9	52.6	62.4	52.0	50.0	56.6	67.0
	65	49.1	47.2	52.6	60.7	50.0	48.2	53.8	63.2	53.1	51.1	57.5	67.9
	70	53.2	51.0	57.0	65.3	53.3	51.0	57.1	65.4	54.5	52.3	58.6	68.6
	75	57.4	54.7	61.6	70.0	57.5	54.8	61.6	70.1	57.6	54.6	61.7	70.0

Note: MIN, MAX: The leaving evaporator temperature is affected by the modulating valve position. The MIN and MAX numbers represent modulating valve position impact on both the leaving evaporator temperature and the leaving reheat coil temperature. MAX represents wide open, MIN represents closed to minimum position.

^(a) The unit might be tripped by Froststat at these conditions due to low leaving evaporator air temperature.



Evaporator Fan Performance

Fan Curve Limits:

-
- THJ60-120 Std Motor, Max 3.0 hp or 1850 rpm
- THJ150 Std Motor, Max 4.55 hp or 1940 rpm
- THJ120 High Static, Max 4.55 hp or 1940 rpm
- THJ180-300 Std Motor, Max 6 hp or 1850 rpm
- THJ300 High static, Max 9.1 hp or 1940 rpm
- Maximum CFM — 480 cfm/ton
- Maximum ESP = 2.0 in-H₂O @ 400 cfm/ton

The fan curve graphs include standard filter and a wet indoor coil. ESP capability is reduced with options based on the accessory table component pressure drop. To determine ESP at rpm/cfm with other options/accessories, select intersection point of the RPM vs CFM and then reduce ESP shown in graph by the sum of additional option static pressure drop listed in the fan performance accessory table section.

6 to 10 Ton Units — Downflow

Figure 2. Fan curves — 6 to 10 tons, downflow

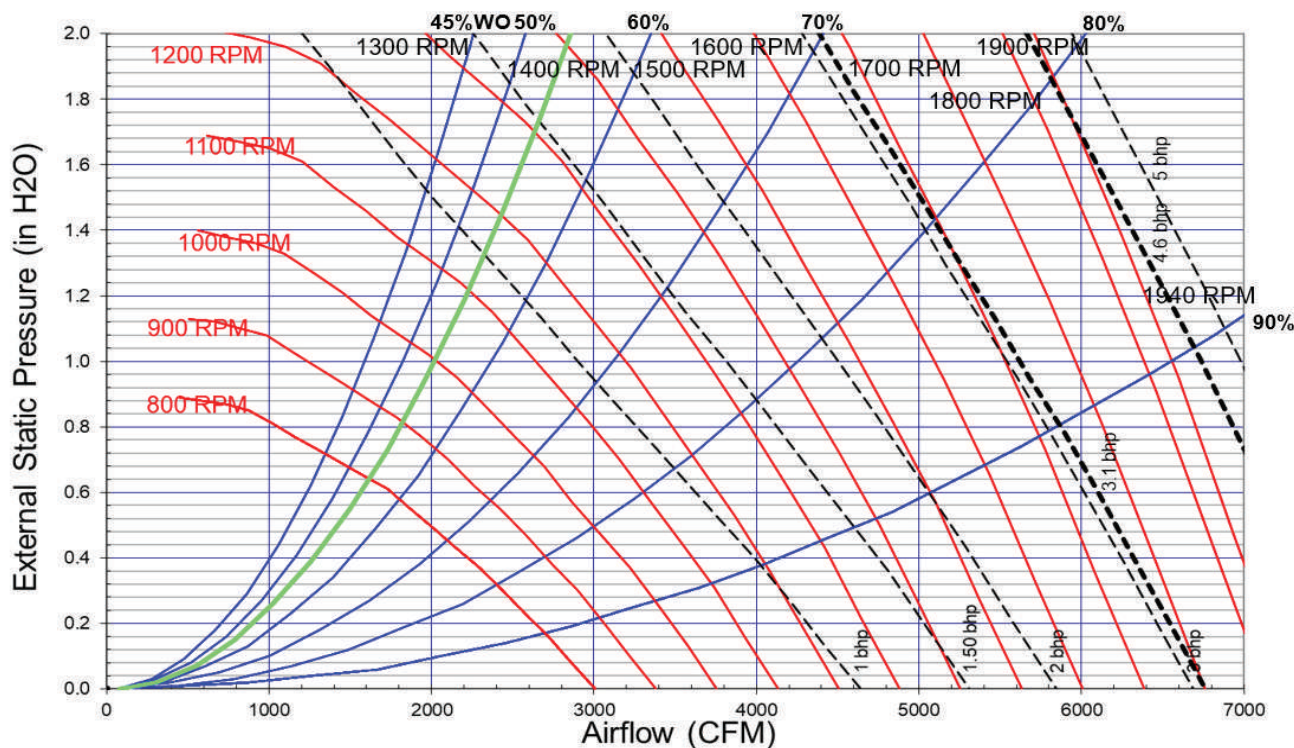


Table 23. Evaporator fan performance - 6 ton (model THJ), downflow

Available External Static Pressure (Inches of Water Gauge)										
CFM	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
1800	549 0.10	612 0.15	667 0.20	719 0.25	766 0.30	809 0.35	849 0.41	891 0.47	931 0.53	969 0.59
1920	577 0.12	638 0.16	691 0.22	741 0.27	787 0.32	830 0.38	870 0.44	907 0.50	946 0.56	984 0.62
2040	606 0.13	663 0.18	716 0.24	763 0.29	809 0.35	851 0.41	891 0.47	928 0.53	963 0.60	999 0.66
2160	634 0.15	689 0.20	741 0.26	787 0.32	831 0.38	873 0.44	912 0.50	948 0.57	983 0.63	1017 0.70

Table 23. Evaporator fan performance - 6 ton (model THJ), downflow (continued)

Available External Static Pressure (Inches of Water Gauge)										
	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
CFM	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
2280	663 0.17	716 0.23	766 0.29	811 0.35	853 0.41	894 0.47	933 0.54	969 0.61	1004 0.67	1037 0.74
2400	693 0.19	743 0.25	791 0.31	836 0.38	877 0.44	916 0.51	955 0.58	991 0.65	1025 0.72	1058 0.79
2520	723 0.22	771 0.28	817 0.34	861 0.41	901 0.48	939 0.55	977 0.62	1012 0.69	1046 0.76	1079 0.83
2640	753 0.24	799 0.31	843 0.37	886 0.44	925 0.51	963 0.59	999 0.66	1034 0.73	1068 0.81	1100 0.88
2760	783 0.27	827 0.34	869 0.41	911 0.48	950 0.55	987 0.63	1022 0.70	1056 0.78	1089 0.86	1121 0.94
2880	813 0.30	855 0.37	897 0.44	937 0.52	975 0.59	1011 0.67	1046 0.75	1078 0.83	1111 0.91	1143 0.99
Available External Static Pressure (Inches of Water Gauge)										
	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
CFM	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
1800	1005 0.65	1040 0.71	1074 0.78	1106 0.84	1137 0.91	1167 0.98	1197 1.04	1225 1.11	1253 1.19	1280 1.26
1920	1020 0.69	1055 0.75	1088 0.82	1121 0.89	1152 0.96	1182 1.03	1211 1.1	1240 1.17	1267 1.24	1294 1.32
2040	1035 0.73	1070 0.8	1103 0.86	1135 0.93	1166 1.01	1196 1.08	1226 1.15	1254 1.23	1282 1.3	1309 1.38
2160	1050 0.77	1085 0.84	1118 0.91	1150 0.98	1181 1.06	1211 1.13	1240 1.21	1269 1.28	1296 1.36	1323 1.44
2280	1069 0.81	1100 0.89	1133 0.96	1165 1.04	1196 1.11	1226 1.19	1255 1.27	1283 1.35	1311 1.43	1338 1.51
2400	1089 0.86	1119 0.94	1149 1.01	1180 1.09	1211 1.17	1241 1.25	1270 1.33	1298 1.41	1326 1.49	1352 1.57
2520	1110 0.91	1140 0.99	1169 1.07	1197 1.15	1226 1.23	1256 1.31	1285 1.39	1313 1.47	1340 1.56	1367 1.64
2640	1131 0.96	1161 1.04	1189 1.12	1217 1.2	1244 1.29	1271 1.37	1300 1.46	1328 1.54	1355 1.63	1382 1.72
2760	1152 1.02	1182 1.1	1210 1.18	1238 1.26	1265 1.35	1291 1.44	1316 1.52	1343 1.61	1371 1.7	1397 1.79
2880	1173 1.07	1203 1.16	1231 1.24	1259 1.33	1285 1.41	1311 1.5	1336 1.59	1361 1.68	1386 1.77	1412 1.87

Notes:

1. Available External Static Pressure is the static pressure difference between the return duct and the supply duct plus the static pressure drop caused by accessories and options.
2. For direct drive evaporator fan speed (rpm), refer to the applicable table in the fan performance section.
3. Data includes pressure drop due to standard filters and wet coils. No accessories or options are included in pressure drop data.
4. To determine static pressure drop due to other options/accessories, refer to the applicable table in the fan performance section.
5. Direct drive fan motor heat is negligible.
6. Factory supplied motors, in commercial equipment, are definite purpose motors, specifically designed and tested to operate reliably and continuously at all cataloged conditions. Using the full horsepower range of our fan motors as shown in our tabular data will not result in nuisance tripping or premature motor failure. Our product's warranty will not be affected.

Table 24. Evaporator fan performance - 7.5 ton (model THJ), downflow

Available External Static Pressure (Inches of Water Gauge)										
	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
CFM	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
2250	656 0.17	709 0.22	759 0.28	805 0.34	848 0.40	889 0.47	928 0.53	964 0.60	999 0.66	1032 0.73
2400	693 0.19	743 0.25	791 0.31	836 0.38	877 0.44	916 0.51	955 0.58	991 0.65	1025 0.72	1058 0.79
2550	730 0.22	778 0.29	823 0.35	867 0.42	907 0.49	945 0.56	982 0.63	1018 0.70	1052 0.77	1084 0.85
2700	768 0.26	813 0.32	856 0.39	898 0.46	938 0.53	975 0.61	1010 0.68	1045 0.76	1079 0.83	1111 0.91
2850	806 0.30	848 0.36	890 0.43	930 0.51	969 0.58	1005 0.66	1040 0.74	1073 0.82	1106 0.90	1138 0.98
3000	844 0.34	884 0.41	924 0.48	963 0.56	1000 0.64	1036 0.72	1070 0.80	1102 0.88	1134 0.96	1165 1.05
3150	882 0.38	920 0.46	959 0.53	996 0.61	1032 0.69	1067 0.78	1100 0.86	1132 0.95	1162 1.03	1193 1.12
3300	920 0.43	957 0.51	994 0.59	1030 0.67	1065 0.76	1099 0.84	1131 0.93	1163 1.02	1192 1.11	1221 1.20
3600	997 0.55	1031 0.63	1065 0.71	1099 0.80	1131 0.90	1163 0.99	1194 1.08	1225 1.18	1254 1.27	1282 1.37
Available External Static Pressure (Inches of Water Gauge)										
	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
CFM	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
2250	1064 0.8	1096 0.88	1130 0.95	1161 1.02	1192 1.1	1222 1.17	1251 1.25	1280 1.33	1307 1.41	1334 1.49
2400	1089 0.86	1119 0.94	1149 1.01	1180 1.09	1211 1.17	1241 1.25	1270 1.33	1298 1.41	1326 1.49	1352 1.57
2550	1115 0.92	1145 1	1174 1.08	1202 1.16	1230 1.24	1260 1.32	1289 1.41	1317 1.49	1344 1.58	1371 1.66
2700	1141 0.99	1171 1.07	1200 1.15	1227 1.23	1254 1.32	1280 1.4	1308 1.49	1336 1.58	1363 1.66	1390 1.75
2850	1168 1.06	1197 1.14	1226 1.23	1253 1.31	1280 1.4	1306 1.49	1331 1.57	1356 1.66	1382 1.76	1409 1.85
3000	1195 1.13	1224 1.22	1252 1.3	1280 1.39	1306 1.48	1332 1.57	1357 1.67	1381 1.76	1405 1.85	1428 1.95
3150	1223 1.21	1251 1.3	1279 1.39	1306 1.48	1332 1.57	1358 1.67	1383 1.76	1407 1.86	1431 1.95	1454 2.05
3300	1250 1.29	1279 1.38	1306 1.48	1333 1.57	1359 1.67	1384 1.76	1409 1.86	1433 1.96	1457 2.06	1480 2.16



Evaporator Fan Performance

Table 24. Evaporator fan performance - 7.5 ton (model THJ), downflow (continued)

Available External Static Pressure (Inches of Water Gauge)										
CFM	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
3600	1309 1.47	1335 1.57	1361 1.67	1388 1.77	1413 1.87	1438 1.97	1462 2.08	1486 2.18	1509 2.28	1532 2.39

Notes:

1. Available External Static Pressure is the static pressure difference between the return duct and the supply duct plus the static pressure drop caused by accessories and options.
2. For direct drive evaporator fan speed (rpm), refer to the applicable table in the fan performance section.
3. Data includes pressure drop due to standard filters and wet coils. No accessories or options are included in pressure drop data.
4. To determine static pressure drop due to other options/accessories, refer to the applicable table in the fan performance section.
5. Direct drive fan motor heat is negligible.
6. Factory supplied motors, in commercial equipment, are definite purpose motors, specifically designed and tested to operate reliably and continuously at all cataloged conditions. Using the full horsepower range of our fan motors as shown in our tabular data will not result in nuisance tripping or premature motor failure. Our product's warranty will not be affected.

Table 25. Evaporator fan performance - 8.5 ton (model THJ), downflow

Available External Static Pressure (Inches of Water Gauge)										
CFM	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
2550	730 0.22	778 0.29	823 0.35	867 0.42	907 0.49	945 0.56	982 0.63	1018 0.70	1052 0.77	1084 0.85
2720	773 0.26	817 0.33	860 0.40	903 0.47	942 0.54	979 0.61	1014 0.69	1049 0.76	1082 0.84	1114 0.92
2890	816 0.31	858 0.37	899 0.45	939 0.52	977 0.60	1013 0.67	1048 0.75	1080 0.83	1113 0.91	1145 0.99
3060	859 0.36	898 0.43	938 0.50	976 0.58	1013 0.66	1048 0.74	1082 0.82	1114 0.91	1145 0.99	1176 1.08
3230	902 0.41	940 0.48	977 0.56	1014 0.64	1050 0.73	1084 0.81	1117 0.90	1148 0.98	1178 1.07	1208 1.16
3400	946 0.47	982 0.55	1017 0.63	1052 0.71	1086 0.80	1120 0.89	1152 0.98	1183 1.07	1213 1.16	1241 1.25
3570	989 0.54	1024 0.62	1057 0.70	1092 0.79	1124 0.88	1157 0.97	1188 1.06	1218 1.16	1247 1.25	1275 1.35
3740	1033 0.61	1066 0.69	1098 0.78	1131 0.87	1163 0.97	1193 1.06	1224 1.16	1254 1.26	1283 1.35	1310 1.46
4080	1121 0.77	1152 0.86	1181 0.96	1211 1.06	1241 1.16	1270 1.26	1298 1.36	1326 1.47	1354 1.57	1381 1.68
Available External Static Pressure (Inches of Water Gauge)										
CFM	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
2550	1115 0.92	1145 1	1174 1.08	1202 1.16	1230 1.24	1260 1.32	1289 1.41	1317 1.49	1344 1.58	1371 1.66
2720	1145 1	1175 1.08	1203 1.16	1231 1.24	1258 1.33	1284 1.41	1310 1.5	1338 1.59	1366 1.68	1392 1.77
2890	1175 1.08	1205 1.16	1233 1.25	1260 1.33	1287 1.42	1313 1.51	1338 1.6	1362 1.69	1387 1.78	1414 1.87
3060	1206 1.16	1235 1.25	1263 1.34	1290 1.43	1317 1.52	1342 1.61	1367 1.7	1391 1.8	1415 1.89	1439 1.99
3230	1237 1.25	1266 1.34	1294 1.44	1320 1.53	1347 1.62	1372 1.72	1397 1.81	1421 1.91	1444 2.01	1468 2.11
3400	1269 1.35	1297 1.44	1325 1.54	1351 1.63	1377 1.73	1402 1.83	1427 1.93	1451 2.03	1474 2.13	1497 2.23
3570	1303 1.45	1329 1.55	1356 1.65	1382 1.75	1408 1.85	1433 1.95	1457 2.05	1481 2.16	1504 2.26	1527 2.37
3740	1337 1.56	1363 1.66	1388 1.76	1414 1.87	1439 1.97	1464 2.08	1488 2.18	1511 2.29	1534 2.4	1557 2.51
4080	1407 1.79	1432 1.9	1456 2.01	1480 2.12	1503 2.24	1526 2.35	1550 2.46	1573 2.58	1596 2.69	1618 2.81

Notes:

1. Available External Static Pressure is the static pressure difference between the return duct and the supply duct plus the static pressure drop caused by accessories and options.
2. For direct drive evaporator fan speed (rpm), refer to the applicable table in the fan performance section.
3. Data includes pressure drop due to standard filters and wet coils. No accessories or options are included in pressure drop data.
4. To determine static pressure drop due to other options/accessories, refer to the applicable table in the fan performance section.
5. Direct drive fan motor heat is negligible.
6. Factory supplied motors, in commercial equipment, are definite purpose motors, specifically designed and tested to operate reliably and continuously at all cataloged conditions. Using the full horsepower range of our fan motors as shown in our tabular data will not result in nuisance tripping or premature motor failure. Our product's warranty will not be affected.

Table 26. Evaporator fan performance - 10 ton (model THJ), downflow

Available External Static Pressure (Inches of Water Gauge)										
CFM	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
3-hp Standard Motor										
3000	844 0.34	884 0.41	924 0.48	963 0.56	1000 0.64	1036 0.72	1070 0.80	1102 0.88	1134 0.96	1165 1.05
3200	894 0.40	932 0.47	970 0.55	1007 0.63	1043 0.71	1078 0.80	1111 0.88	1142 0.97	1172 1.06	1202 1.15
3400	946 0.47	982 0.55	1017 0.63	1052 0.71	1086 0.80	1120 0.89	1152 0.98	1183 1.07	1213 1.16	1241 1.25
3600	997 0.55	1031 0.63	1065 0.71	1099 0.80	1131 0.90	1163 0.99	1194 1.08	1225 1.18	1254 1.27	1282 1.37
3800	1048 0.64	1081 0.72	1113 0.81	1145 0.90	1176 1.00	1206 1.09	1237 1.19	1267 1.29	1295 1.39	1322 1.49

Table 26. Evaporator fan performance - 10 ton (model THJ), downflow (continued)

Available External Static Pressure (Inches of Water Gauge)										
CFM	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
3-hp Standard Motor										
4000	1100 0.73	1132 0.82	1162 0.91	1192 1.01	1222 1.11	1251 1.21	1280 1.31	1309 1.42	1337 1.52	1364 1.63
4200	1152 0.84	1182 0.93	1211 1.03	1240 1.13	1269 1.23	1297 1.34	1324 1.44	1352 1.55	1379 1.66	1406 1.77
4400	1204 0.95	1233 1.05	1261 1.15	1288 1.25	1316 1.36	1343 1.47	1370 1.58	1396 1.69	1422 1.81	1448 1.92
4800	1308 1.22	1335 1.33	1361 1.43	1386 1.54	1411 1.66	1437 1.77	1462 1.89	1486 2.01	1510 2.14	1534 2.26
Available External Static Pressure (Inches of Water Gauge)										
CFM	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
3-hp Standard Motor										
3000	1195 1.13	1224 1.22	1252 1.3	1280 1.39	1306 1.48	1332 1.57	1357 1.67	1381 1.76	1405 1.85	1428 1.95
3200	1232 1.24	1260 1.33	1288 1.42	1315 1.51	1341 1.60	1367 1.70	1391 1.79	1416 1.89	1439 1.99	1462 2.09
3400	1269 1.35	1297 1.44	1325 1.54	1351 1.63	1377 1.73	1402 1.83	1427 1.93	1451 2.03	1474 2.13	1497 2.23
3600	1309 1.47	1335 1.57	1361 1.67	1388 1.77	1413 1.87	1438 1.97	1462 2.08	1486 2.18	1509 2.28	1532 2.39
3800	1349 1.60	1375 1.07	1400 1.81	1425 1.91	1450 2.02	1475 2.12	1499 2.23	1522 2.34	1545 2.45	1567 2.56
4000	1390 1.73	1415 1.84	1440 1.95	1464 2.06	1487 2.17	1512 2.28	1535 2.39	1558 2.51	1581 2.62	1603 2.73
4200	1431 1.88	1456 1.99	1481 2.11	1504 2.22	1527 2.34	1550 2.45	1572 2.57	1595 2.69	1618 2.80	1640 2.92
4400	1473 2.04	1498 2.15	1522 2.27	1545 2.39	1568 2.51	1590 2.63	1612 2.75	1633 2.87	1655 3.00	- -
4800	1559 2.38	1582 2.51	1605 2.63	1628 2.76	1650 2.89	1671 3.02	- -	- -	- -	- -

Notes:

1. Available External Static Pressure is the static pressure difference between the return duct and the supply duct plus the static pressure drop caused by accessories and options.
2. For direct drive evaporator fan speed (rpm), refer to the applicable table in the fan performance section.
3. Data includes pressure drop due to standard filters and wet coils. No accessories or options are included in pressure drop data.
4. To determine static pressure drop due to other options/accessories, refer to the applicable table in the fan performance section.
5. Direct drive fan motor heat is negligible.
6. Factory supplied motors, in commercial equipment, are definite purpose motors, specifically designed and tested to operate reliably and continuously at all cataloged conditions. Using the full horsepower range of our fan motors as shown in our tabular data will not result in nuisance tripping or premature motor failure. Our product's warranty will not be affected.

Table 27. Evaporator fan performance - 10 ton (model THJ), downflow, oversized/high static motor option

Available External Static Pressure (Inches of Water Gauge)										
CFM	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
5-hp Oversized Motor										
3000	816 0.31	859 0.38	900 0.46	941 0.53	980 0.62	1018 0.70	1054 0.78	1089 0.87	1123 0.96	1158 1.05
3200	865 0.37	905 0.44	944 0.52	982 0.60	1020 0.69	1057 0.77	1092 0.86	1126 0.95	1159 1.04	1191 1.14
3400	913 0.43	952 0.51	989 0.59	1026 0.68	1061 0.76	1097 0.86	1130 0.95	1163 1.04	1195 1.14	1226 1.24
3600	962 0.50	1000 0.58	1034 0.67	1069 0.76	1103 0.85	1137 0.95	1170 1.04	1202 1.14	1232 1.24	1263 1.34
3800	1012 0.58	1047 0.67	1081 0.75	1114 0.85	1146 0.94	1178 1.04	1210 1.14	1241 1.25	1271 1.35	1300 1.46
4000	1061 0.66	1095 0.76	1128 0.85	1158 0.95	1190 1.05	1220 1.15	1251 1.25	1281 1.36	1310 1.47	1339 1.58
4200	1111 0.76	1144 0.85	1175 0.95	1204 1.05	1234 1.16	1264 1.26	1293 1.37	1322 1.48	1350 1.60	1378 1.71
4400	1160 0.86	1192 0.96	1222 1.07	1251 1.17	1279 1.28	1308 1.39	1335 1.50	1363 1.61	1391 1.73	1418 1.85
4600	1210 0.97	1241 1.08	1270 1.19	1298 1.30	1325 1.41	1352 1.52	1379 1.63	1405 1.75	1432 1.87	1458 2.00
4800	1260 1.10	1290 1.21	1318 1.32	1345 1.43	1371 1.55	1397 1.66	1423 1.78	1448 1.90	1474 2.03	1499 2.15
Available External Static Pressure (Inches of Water Gauge)										
CFM	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
5-hp Oversized Motor										
3000	1192 1.15	1224 1.24	1253 1.33	1281 1.43	1308 1.52	1334 1.62	1360 1.71	1384 1.81	1409 1.91	1432 2.01
3200	1223 1.24	1255 1.34	1287 1.44	1315 1.54	1342 1.64	1368 1.74	1393 1.84	1418 1.94	1442 2.05	1466 2.15
3400	1257 1.34	1286 1.44	1317 1.55	1347 1.65	1376 1.76	1402 1.87	1428 1.97	1452 2.08	1476 2.19	1499 2.29
3600	1292 1.45	1321 1.55	1349 1.66	1378 1.77	1407 1.88	1435 2	1462 2.11	1486 2.22	1510 2.34	1533 2.45
3800	1329 1.56	1357 1.67	1384 1.78	1411 1.89	1438 2.01	1466 2.13	1493 2.25	1519 2.37	1545 2.49	1568 2.61
4000	1366 1.69	1393 1.8	1420 1.92	1447 2.03	1473 2.15	1498 2.27	1524 2.39	1550 2.52	1576 2.64	1600 2.77
4200	1405 1.83	1431 1.94	1457 2.06	1483 2.18	1508 2.3	1533 2.42	1557 2.55	1581 2.67	1607 2.8	1631 2.93
4400	1444 1.97	1470 2.09	1495 2.21	1520 2.34	1545 2.46	1569 2.59	1593 2.71	1616 2.84	1639 2.97	1663 3.1



Evaporator Fan Performance

Table 27. Evaporator fan performance - 10 ton (model THJ), downflow, oversized/high static motor option (continued)

Available External Static Pressure (Inches of Water Gauge)										
CFM	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
5-hp Oversized Motor										
4600	1484 2.12	1509 2.25	1534 2.38	1558 2.5	1582 2.63	1605 2.76	1629 2.89	1652 3.02	1675 3.16	1697 3.29
4800	1525 2.28	1549 2.42	1573 2.55	1597 2.68	1620 2.81	1643 2.94	1666 3.08	1688 3.21	1711 3.35	1733 3.49

Notes:

1. Available External Static Pressure is the static pressure difference between the return duct and the supply duct plus the static pressure drop caused by accessories and options.
2. For direct drive evaporator fan speed (rpm), refer to the applicable table in the fan performance section.
3. Data includes pressure drop due to standard filters and wet coils. No accessories or options are included in pressure drop data.
4. To determine static drive pressure drop due to other options/accessories, refer to the applicable table in the fan performance section.
5. Direct drive fan motor heat is negligible.
6. Factory supplied motors, in commercial equipment, are definite purpose motors, specifically designed and tested to operate reliably and continuously at all cataloged conditions. Using the full horsepower range of our fan motors as shown in our tabular data will not result in nuisance tripping or premature motor failure. Our product's warranty will not be affected.

6 to 10 Ton Units — Horizontal

Figure 3. Fan curves — 6 to 10 tons, horizontal

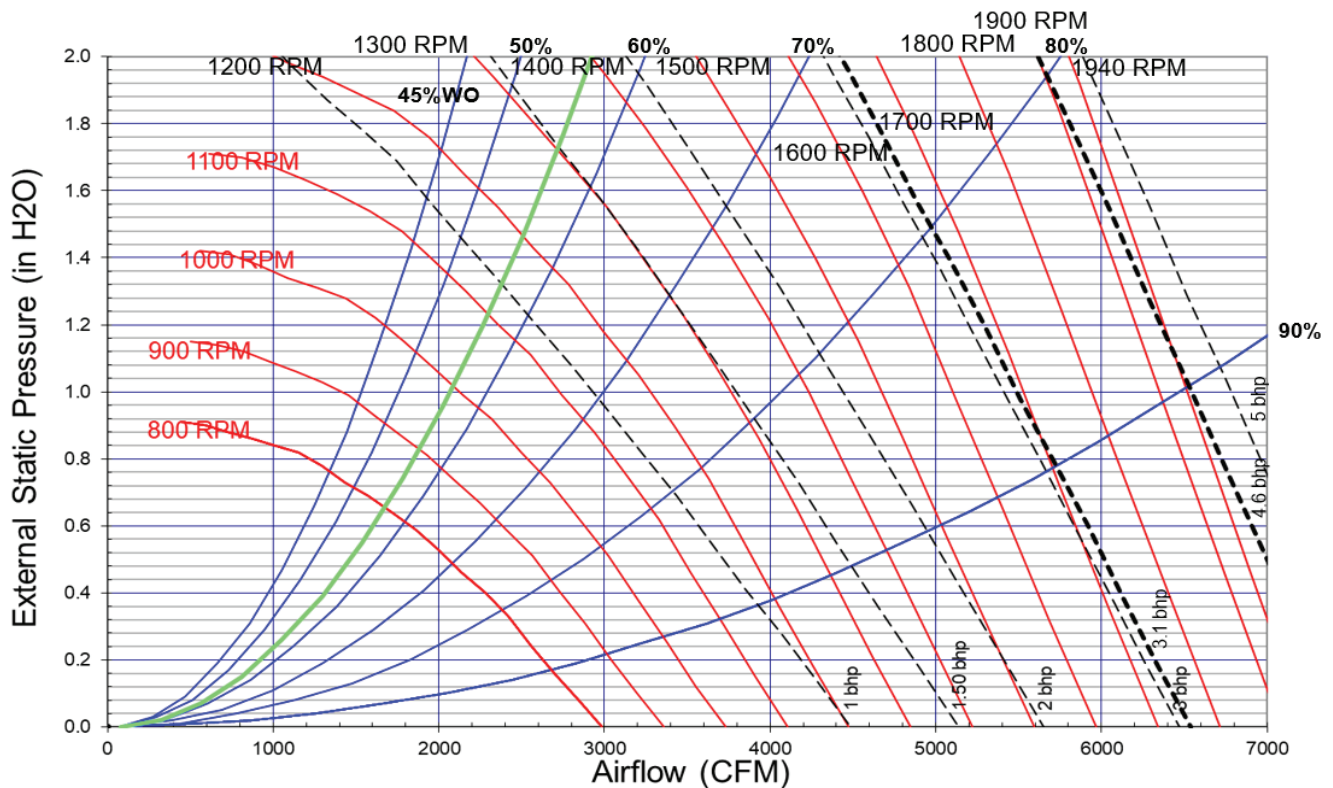


Table 28. Evaporator fan performance - 6 ton (model THJ), horizontal

Available External Static Pressure (Inches of Water Gauge)										
CFM	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
1800	550 0.11	606 0.15	658 0.19	708 0.24	754 0.29	797 0.35	839 0.40	878 0.46	915 0.52	951 0.58
1920	579 0.12	632 0.17	682 0.22	730 0.27	775 0.32	817 0.37	858 0.43	896 0.49	933 0.55	968 0.61
2040	608 0.14	659 0.19	707 0.24	753 0.29	797 0.35	838 0.40	876 0.46	915 0.52	951 0.59	986 0.65
2160	638 0.16	687 0.21	732 0.26	776 0.32	819 0.37	859 0.43	897 0.49	933 0.56	970 0.62	1004 0.69
2280	667 0.18	715 0.24	758 0.29	801 0.35	842 0.41	881 0.47	919 0.53	954 0.59	988 0.66	1023 0.73
2400	697 0.21	744 0.26	785 0.32	825 0.38	865 0.44	904 0.50	940 0.57	975 0.63	1009 0.70	1041 0.77
2520	727 0.24	772 0.29	812 0.35	851 0.41	889 0.48	926 0.54	963 0.61	997 0.68	1030 0.75	1061 0.82

Table 28. Evaporator fan performance - 6 ton (model THJ), horizontal (continued)

Available External Static Pressure (Inches of Water Gauge)										
	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
CFM	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
2640	758 0.26	801 0.33	840 0.39	877 0.45	914 0.52	950 0.58	985 0.65	1019 0.72	1052 0.79	1083 0.87
2760	788 0.30	830 0.36	868 0.42	904 0.49	939 0.56	974 0.63	1008 0.70	1041 0.77	1073 0.84	1104 0.92
2880	818 0.33	859 0.40	896 0.46	930 0.53	965 0.60	998 0.67	1032 0.75	1064 0.82	1096 0.90	1126 0.97
Available External Static Pressure (Inches of Water Gauge)										
	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
CFM	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
1800	985 0.64	1018 0.7	1050 0.76	1080 0.83	1109 0.89	1138 0.96	1165 1.03	1193 1.1	1223 1.17	1251 1.25
1920	1002 0.68	1034 0.74	1066 0.81	1096 0.87	1126 0.94	1154 1.01	1181 1.08	1208 1.15	1234 1.22	1260 1.29
2040	1019 0.72	1051 0.78	1083 0.85	1113 0.92	1142 0.99	1170 1.06	1198 1.13	1224 1.21	1250 1.28	1276 1.35
2160	1037 0.76	1069 0.83	1099 0.9	1129 0.97	1158 1.04	1187 1.11	1214 1.19	1241 1.26	1267 1.34	1292 1.42
2280	1056 0.8	1087 0.87	1117 0.94	1147 1.02	1175 1.09	1203 1.17	1231 1.25	1257 1.32	1283 1.4	1308 1.48
2400	1074 0.84	1105 0.92	1135 0.99	1165 1.07	1193 1.15	1220 1.23	1247 1.31	1274 1.39	1300 1.47	1325 1.55
2520	1093 0.89	1124 0.97	1154 1.05	1183 1.12	1211 1.2	1238 1.29	1265 1.37	1291 1.45	1316 1.53	1341 1.62
2640	1113 0.94	1142 1.02	1172 1.1	1201 1.18	1229 1.26	1256 1.35	1283 1.43	1308 1.52	1333 1.6	1358 1.69
2760	1134 1	1163 1.08	1191 1.16	1220 1.24	1248 1.32	1275 1.41	1301 1.49	1326 1.58	1351 1.67	1376 1.76
2880	1156 1.05	1184 1.13	1211 1.22	1238 1.3	1266 1.39	1293 1.47	1319 1.56	1345 1.65	1369 1.74	1394 1.83

Notes:

1. Available External Static Pressure is the static pressure difference between the return duct and the supply duct plus the static pressure drop caused by accessories and options.
2. For direct drive evaporator fan speed (rpm), refer to the applicable table in the fan performance section.
3. Data includes pressure drop due to standard filters and wet coils. No accessories or options are included in pressure drop data.
4. To determine static pressure drop due to other options/accessories, refer to the applicable table in the fan performance section.
5. Direct drive fan motor heat is negligible.
6. Factory supplied motors, in commercial equipment, are definite purpose motors, specifically designed and tested to operate reliably and continuously at all cataloged conditions. Using the full horsepower range of our fan motors as shown in our tabular data will not result in nuisance tripping or premature motor failure. Our product's warranty will not be affected.

Table 29. Evaporator fan performance - 7.5 ton (model THJ), horizontal

Available External Static Pressure (Inches of Water Gauge)										
	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
CFM	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
2250	660 0.18	708 0.23	752 0.28	795 0.34	836 0.40	876 0.46	913 0.52	949 0.58	984 0.65	1018 0.72
2400	697 0.21	744 0.26	785 0.32	825 0.38	865 0.44	904 0.50	940 0.57	975 0.63	1009 0.70	1041 0.77
2550	735 0.24	779 0.30	819 0.36	857 0.42	895 0.49	932 0.55	968 0.62	1003 0.69	1035 0.76	1067 0.83
2700	773 0.28	816 0.34	854 0.40	890 0.47	926 0.54	962 0.60	996 0.67	1030 0.74	1062 0.82	1094 0.89
2850	811 0.32	852 0.39	889 0.45	924 0.52	958 0.59	992 0.66	1026 0.73	1058 0.81	1090 0.88	1121 0.96
3000	849 0.37	889 0.44	925 0.51	958 0.58	991 0.65	1023 0.72	1056 0.80	1087 0.87	1118 0.95	1148 1.03
3150	887 0.42	926 0.49	961 0.56	994 0.64	1025 0.71	1056 0.79	1087 0.87	1117 0.95	1147 1.03	1176 1.11
3300	926 0.47	963 0.55	997 0.63	1029 0.70	1059 0.78	1089 0.86	1118 0.94	1148 1.02	1177 1.11	1205 1.19
3600	1003 0.60	1038 0.68	1070 0.77	1101 0.85	1129 0.93	1156 1.01	1184 1.10	1211 1.19	1238 1.28	1265 1.37
Available External Static Pressure (Inches of Water Gauge)										
	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
CFM	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
2250	1051 0.79	1083 0.86	1113 0.93	1142 1.01	1171 1.08	1199 1.16	1226 1.23	1253 1.31	1279 1.39	1304 1.46
2400	1074 0.84	1105 0.92	1135 0.99	1165 1.07	1193 1.15	1220 1.23	1247 1.31	1274 1.39	1300 1.47	1325 1.55
2550	1097 0.9	1128 0.98	1158 1.06	1187 1.14	1215 1.22	1243 1.3	1269 1.38	1295 1.47	1320 1.55	1345 1.63
2700	1123 0.97	1152 1.05	1182 1.13	1210 1.21	1238 1.29	1265 1.38	1292 1.46	1317 1.55	1342 1.64	1367 1.72
2850	1150 1.04	1179 1.12	1206 1.2	1234 1.28	1261 1.37	1288 1.46	1315 1.55	1340 1.63	1365 1.72	1389 1.82
3000	1177 1.11	1206 1.2	1233 1.28	1259 1.37	1285 1.45	1312 1.54	1338 1.63	1363 1.72	1388 1.82	1412 1.91
3150	1205 1.19	1233 1.28	1260 1.36	1286 1.45	1311 1.54	1336 1.63	1361 1.72	1386 1.82	1411 1.92	1435 2.01
3300	1233 1.28	1260 1.36	1287 1.45	1313 1.54	1338 1.64	1363 1.73	1386 1.82	1410 1.92	1434 2.02	1458 2.12



Evaporator Fan Performance

Table 29. Evaporator fan performance - 7.5 ton (model THJ), horizontal (continued)

Available External Static Pressure (Inches of Water Gauge)										
CFM	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
3600	1291 1.46	1317 1.55	1343 1.65	1368 1.74	1393 1.84	1417 1.94	1440 2.04	1463 2.14	1486 2.24	1507 2.34

Notes:

1. Available External Static Pressure is the static pressure difference between the return duct and the supply duct plus the static pressure drop caused by accessories and options.
2. For direct drive evaporator fan speed (rpm), refer to the applicable table in the fan performance section.
3. Data includes pressure drop due to standard filters and wet coils. No accessories or options are included in pressure drop data.
4. To determine static pressure drop due to other options/accessories, refer to the applicable table in the fan performance section.
5. Direct drive fan motor heat is negligible.
6. Factory supplied motors, in commercial equipment, are definite purpose motors, specifically designed and tested to operate reliably and continuously at all cataloged conditions. Using the full horsepower range of our fan motors as shown in our tabular data will not result in nuisance tripping or premature motor failure. Our product's warranty will not be affected.

Table 30. Evaporator fan performance - 8.5 ton (model THJ), horizontal

Available External Static Pressure (Inches of Water Gauge)										
CFM	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
2550	735 0.24	779 0.30	819 0.36	857 0.42	895 0.49	932 0.55	968 0.62	1003 0.69	1035 0.76	1067 0.83
2720	778 0.29	820 0.35	859 0.41	895 0.48	930 0.54	966 0.61	1000 0.68	1034 0.75	1066 0.83	1097 0.90
2890	821 0.33	862 0.40	899 0.47	933 0.53	967 0.61	1001 0.68	1034 0.75	1066 0.82	1097 0.90	1128 0.98
3060	864 0.39	903 0.46	940 0.53	972 0.60	1004 0.67	1036 0.75	1068 0.82	1099 0.90	1129 0.98	1159 1.06
3230	908 0.45	945 0.52	980 0.60	1012 0.67	1043 0.75	1073 0.83	1103 0.91	1133 0.99	1163 1.07	1191 1.15
3400	952 0.51	988 0.59	1021 0.67	1053 0.75	1082 0.83	1111 0.91	1140 0.99	1168 1.08	1197 1.16	1225 1.25
3570	996 0.59	1030 0.67	1063 0.75	1094 0.83	1122 0.92	1150 1.00	1177 1.08	1205 1.17	1232 1.26	1259 1.35
3740	1040 0.67	1073 0.75	1104 0.84	1134 0.92	1163 1.01	1189 1.10	1215 1.18	1242 1.28	1267 1.37	1293 1.46
4080	1128 0.85	1159 0.94	1189 1.03	1217 1.13	1244 1.22	1269 1.32	1294 1.41	1318 1.50	1342 1.60	1366 1.71

Available External Static Pressure (Inches of Water Gauge)

CFM	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
2550	1097 0.9	1128 0.98	1158 1.06	1187 1.14	1215 1.22	1243 1.3	1269 1.38	1295 1.47	1320 1.55	1345 1.63
2720	1127 0.98	1156 1.06	1185 1.14	1214 1.22	1241 1.3	1268 1.39	1295 1.47	1320 1.56	1345 1.65	1370 1.74
2890	1157 1.06	1186 1.14	1213 1.22	1240 1.31	1268 1.39	1295 1.48	1321 1.57	1346 1.66	1371 1.75	1395 1.84
3060	1188 1.14	1216 1.23	1243 1.31	1270 1.4	1295 1.49	1321 1.58	1347 1.67	1372 1.76	1397 1.86	1421 1.95
3230	1220 1.24	1247 1.32	1274 1.41	1300 1.5	1326 1.59	1350 1.68	1374 1.78	1399 1.87	1423 1.97	1447 2.07
3400	1252 1.33	1279 1.42	1305 1.52	1331 1.61	1356 1.7	1380 1.8	1404 1.89	1427 1.99	1450 2.09	1474 2.19
3570	1285 1.44	1311 1.53	1337 1.63	1362 1.72	1387 1.82	1411 1.92	1435 2.02	1458 2.12	1480 2.22	1502 2.32
3740	1319 1.56	1345 1.65	1369 1.75	1394 1.84	1418 1.94	1442 2.04	1466 2.15	1488 2.25	1511 2.35	1532 2.46
4080	1389 1.81	1413 1.91	1437 2.01	1460 2.11	1483 2.22	1506 2.32	1528 2.43	1551 2.54	1573 2.65	1594 2.76

Notes:

1. Available External Static Pressure is the static pressure difference between the return duct and the supply duct plus the static pressure drop caused by accessories and options.
2. For direct drive evaporator fan speed (rpm), refer to the applicable table in the fan performance section.
3. Data includes pressure drop due to standard filters and wet coils. No accessories or options are included in pressure drop data.
4. To determine static pressure drop due to other options/accessories, refer to the applicable table in the fan performance section.
5. Direct drive fan motor heat is negligible.
6. Factory supplied motors, in commercial equipment, are definite purpose motors, specifically designed and tested to operate reliably and continuously at all cataloged conditions. Using the full horsepower range of our fan motors as shown in our tabular data will not result in nuisance tripping or premature motor failure. Our product's warranty will not be affected.

Table 31. Evaporator fan performance - 10 ton (model THJ), horizontal

Available External Static Pressure (Inches of Water Gauge)										
CFM	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
3-hp Standard Motor										
3000	849 0.37	889 0.44	925 0.51	958 0.58	991 0.65	1023 0.72	1056 0.80	1087 0.87	1118 0.95	1148 1.03
3200	900 0.44	938 0.51	973 0.58	1005 0.66	1036 0.73	1067 0.81	1097 0.89	1127 0.97	1157 1.05	1186 1.13
3400	952 0.51	988 0.59	1021 0.67	1053 0.75	1082 0.83	1111 0.91	1140 0.99	1168 1.08	1197 1.16	1225 1.25
3600	1003 0.60	1038 0.68	1070 0.77	1101 0.85	1129 0.93	1156 1.01	1184 1.10	1211 1.19	1238 1.28	1265 1.37
3800	1055 0.70	1088 0.78	1119 0.87	1149 0.96	1177 1.05	1203 1.13	1229 1.22	1255 1.31	1281 1.41	1306 1.50

Table 31. Evaporator fan performance - 10 ton (model THJ), horizontal (continued)

Available External Static Pressure (Inches of Water Gauge)										
CFM	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
3-hp Standard Motor										
4000	1107 0.80	1139 0.89	1169 0.98	1197 1.08	1225 1.17	1250 1.26	1275 1.35	1299 1.45	1324 1.55	1349 1.65
4200	1159 0.92	1190 1.01	1219 1.11	1246 1.21	1273 1.31	1298 1.40	1322 1.50	1345 1.60	1369 1.69	1392 1.80
4400	1211 1.05	1241 1.15	1269 1.25	1295 1.35	1321 1.45	1346 1.55	1370 1.65	1392 1.75	1414 1.86	1437 1.96
4800	1316 1.34	1344 1.45	1370 1.56	1395 1.67	1419 1.78	1442 1.89	1465 2.00	1487 2.11	1508 2.22	1529 2.33
Available External Static Pressure (Inches of Water Gauge)										
CFM	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
3-hp Standard Motor										
3000	1177 1.11	1206 1.2	1233 1.28	1259 1.37	1285 1.45	1312 1.54	1338 1.63	1363 1.72	1388 1.82	1412 1.91
3200	1214 1.22	1242 1.31	1269 1.39	1295 1.48	1320 1.57	1345 1.66	1369 1.76	1394 1.85	1419 1.95	1443 2.05
3400	1252 1.33	1279 1.42	1305 1.52	1331 1.61	1356 1.70	1380 1.80	1404 1.89	1427 1.99	1450 2.09	1474 2.19
3600	1291 1.46	1317 1.55	1343 1.65	1368 1.74	1393 1.84	1417 1.94	1440 2.04	1463 2.14	1486 2.24	1507 2.34
3800	1332 1.60	1357 1.69	1381 1.79	1405 1.89	1430 1.99	1453 2.09	1477 2.19	1499 2.3	1521 2.40	1543 2.51
4000	1373 1.75	1397 1.85	1421 1.95	1444 2.05	1467 2.15	1491 2.25	1514 2.36	1536 2.47	1558 2.58	1579 2.69
4200	1415 1.90	1438 2.01	1462 2.11	1485 2.22	1507 2.32	1529 2.43	1551 2.54	1573 2.65	1595 2.76	1616 2.88
4400	1459 2.07	1481 2.18	1503 2.29	1525 2.40	1547 2.51	1569 2.62	1590 2.73	1611 2.84	1632 2.96	- -
4800	1549 2.44	1569 2.56	1590 2.68	1610 2.80	1630 2.92	1650 3.04	- -	- -	- -	- -

Notes:

1. Available External Static Pressure is the static pressure difference between the return duct and the supply duct plus the static pressure drop caused by accessories and options.
2. For direct drive evaporator fan speed (rpm), refer to the applicable table in the fan performance section.
3. Data includes pressure drop due to standard filters and wet coils. No accessories or options are included in pressure drop data.
4. To determine static pressure drop due to other options/accessories, refer to the applicable table in the fan performance section.
5. Direct drive fan motor heat is negligible.
6. Factory supplied motors, in commercial equipment, are definite purpose motors, specifically designed and tested to operate reliably and continuously at all cataloged conditions. Using the full horsepower range of our fan motors as shown in our tabular data will not result in nuisance tripping or premature motor failure. Our product's warranty will not be affected.

Table 32. Evaporator fan performance - 10 ton (model THJ), oversized/high static motor option

Available External Static Pressure (Inches of Water Gauge)										
CFM	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
5-hp Oversized Motor										
3000	856 0.38	894 0.45	929 0.52	962 0.58	995 0.65	1028 0.73	1060 0.80	1092 0.88	1122 0.95	1152 1.03
3200	908 0.45	944 0.52	978 0.60	1010 0.67	1040 0.74	1071 0.82	1102 0.90	1132 0.98	1162 1.06	1190 1.14
3400	960 0.53	994 0.61	1027 0.69	1058 0.76	1087 0.84	1114 0.92	1145 1.00	1174 1.08	1202 1.17	1230 1.25
3600	1012 0.62	1045 0.70	1076 0.78	1106 0.87	1134 0.95	1161 1.03	1188 1.11	1216 1.20	1244 1.29	1270 1.38
3800	1065 0.72	1096 0.81	1126 0.89	1155 0.98	1182 1.07	1208 1.15	1233 1.24	1259 1.33	1286 1.42	1312 1.51
4000	1117 0.84	1148 0.92	1176 1.01	1204 1.10	1231 1.19	1256 1.28	1280 1.37	1304 1.47	1329 1.56	1355 1.66
4200	1170 0.96	1199 1.05	1227 1.14	1254 1.24	1279 1.33	1304 1.43	1328 1.52	1351 1.62	1373 1.72	1398 1.82
4400	1223 1.09	1251 1.19	1278 1.29	1304 1.38	1328 1.48	1352 1.58	1376 1.68	1398 1.78	1420 1.88	1441 1.98
4600	1276 1.24	1303 1.34	1329 1.44	1354 1.54	1378 1.65	1401 1.75	1424 1.86	1446 1.96	1467 2.06	1488 2.17
4800	1329 1.40	1355 1.50	1380 1.61	1404 1.72	1428 1.82	1450 1.93	1472 2.04	1494 2.15	1514 2.26	1535 2.37
Available External Static Pressure (Inches of Water Gauge)										
CFM	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
5-hp Oversized Motor										
3000	1180 1.11	1208 1.2	1235 1.28	1261 1.37	1286 1.45	1311 1.54	1337 1.63	1362 1.72	1387 1.81	1411 1.91
3200	1218 1.22	1245 1.31	1271 1.39	1297 1.48	1322 1.57	1346 1.66	1370 1.75	1394 1.85	1418 1.94	1442 2.04
3400	1257 1.34	1283 1.43	1309 1.52	1334 1.61	1359 1.7	1383 1.8	1406 1.89	1429 1.99	1451 2.09	1473 2.18
3600	1297 1.47	1322 1.56	1347 1.65	1372 1.75	1396 1.84	1420 1.94	1443 2.04	1465 2.14	1488 2.24	1509 2.34
3800	1337 1.61	1362 1.7	1387 1.8	1411 1.89	1434 1.99	1458 2.1	1480 2.2	1503 2.3	1524 2.41	1546 2.51
4000	1379 1.76	1403 1.86	1427 1.96	1450 2.06	1473 2.16	1496 2.26	1518 2.37	1540 2.47	1562 2.58	1583 2.69
4200	1422 1.92	1445 2.02	1468 2.12	1491 2.23	1513 2.33	1535 2.44	1557 2.55	1578 2.66	1600 2.77	1620 2.88
4400	1465 2.09	1488 2.2	1510 2.31	1532 2.41	1554 2.52	1576 2.63	1597 2.74	1618 2.86	1638 2.97	1659 3.08



Evaporator Fan Performance

Table 32. Evaporator fan performance - 10 ton (model THJ), oversized/high static motor option (continued)

Available External Static Pressure (Inches of Water Gauge)										
CFM	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
5-hp Oversized Motor										
4600	1508 2.28	1531 2.39	1553 2.5	1574 2.61	1595 2.72	1616 2.84	1637 2.95	1658 3.07	1678 3.18	1698 3.3
4800	1555 2.48	1574 2.59	1596 2.7	1617 2.82	1638 2.94	1658 3.06	1678 3.17	1698 3.29	1718 3.41	1737 3.53

Notes:

1. Available External Static Pressure is the static pressure difference between the return duct and the supply duct plus the static pressure drop caused by accessories and options.
2. For direct drive evaporator fan speed (rpm), refer to the applicable table in the fan performance section.
3. Data includes pressure drop due to standard filters and wet coils. No accessories or options are included in pressure drop data.
4. To determine static pressure drop due to other options/accessories, refer to the applicable table in the fan performance section.
5. Direct drive fan motor heat is negligible.
6. Factory supplied motors, in commercial equipment, are definite purpose motors, specifically designed and tested to operate reliably and continuously at all cataloged conditions. Using the full horsepower range of our fan motors as shown in our tabular data will not result in nuisance tripping or premature motor failure. Our product's warranty will not be affected.

12.5 Ton Units — Downflow

Figure 4. Fan curves — 12.5 ton, downflow

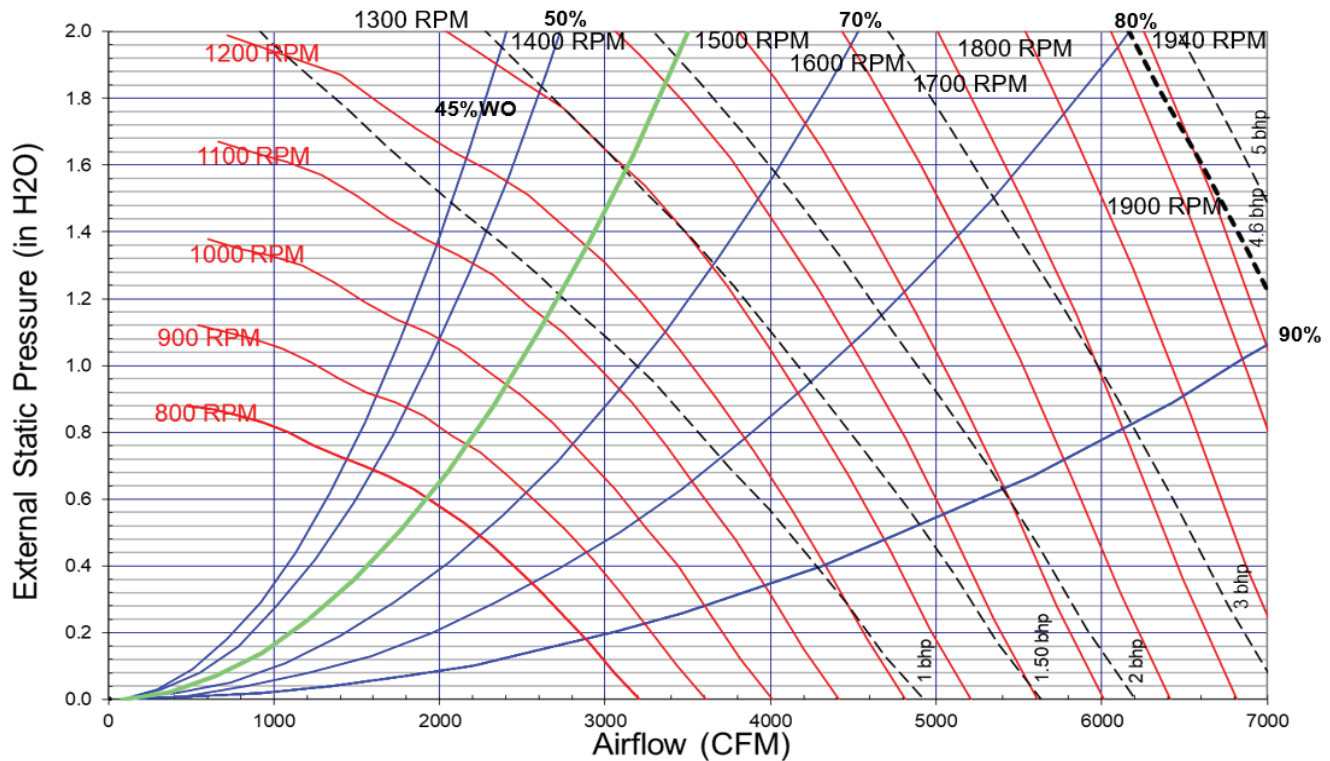


Table 33. Evaporator fan performance - 12.5 ton (model THJ), downflow

Available External Static Pressure (Inches of Water Gauge)										
CFM	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
3750	975 0.52	1008 0.60	1038 0.67	1067 0.75	1099 0.83	1129 0.91	1158 1.00	1186 1.09	1213 1.17	1240 1.26
4000	1036 0.62	1067 0.70	1096 0.78	1124 0.86	1152 0.94	1181 1.03	1209 1.13	1236 1.22	1262 1.31	1288 1.40
4250	1097 0.74	1126 0.82	1154 0.90	1181 0.99	1206 1.07	1233 1.17	1261 1.26	1287 1.36	1313 1.46	1337 1.55
4500	1157 0.86	1186 0.95	1213 1.04	1238 1.13	1263 1.22	1287 1.31	1313 1.41	1339 1.51	1363 1.62	1388 1.72
4750	1218 1.00	1246 1.10	1271 1.19	1296 1.28	1320 1.38	1343 1.47	1366 1.58	1391 1.68	1415 1.79	1438 1.90
5000	1279 1.16	1306 1.26	1331 1.36	1354 1.45	1377 1.55	1399 1.65	1421 1.76	1444 1.86	1467 1.98	1490 2.09
5250	1340 1.33	1367 1.44	1390 1.54	1413 1.64	1435 1.75	1456 1.85	1477 1.96	1498 2.06	1520 2.18	1542 2.30
5500	1401 1.51	1427 1.63	1450 1.74	1472 1.84	1493 1.95	1514 2.06	1534 2.17	1554 2.28	1574 2.40	1595 2.52

Table 33. Evaporator fan performance - 12.5 ton (model THJ), downflow (continued)

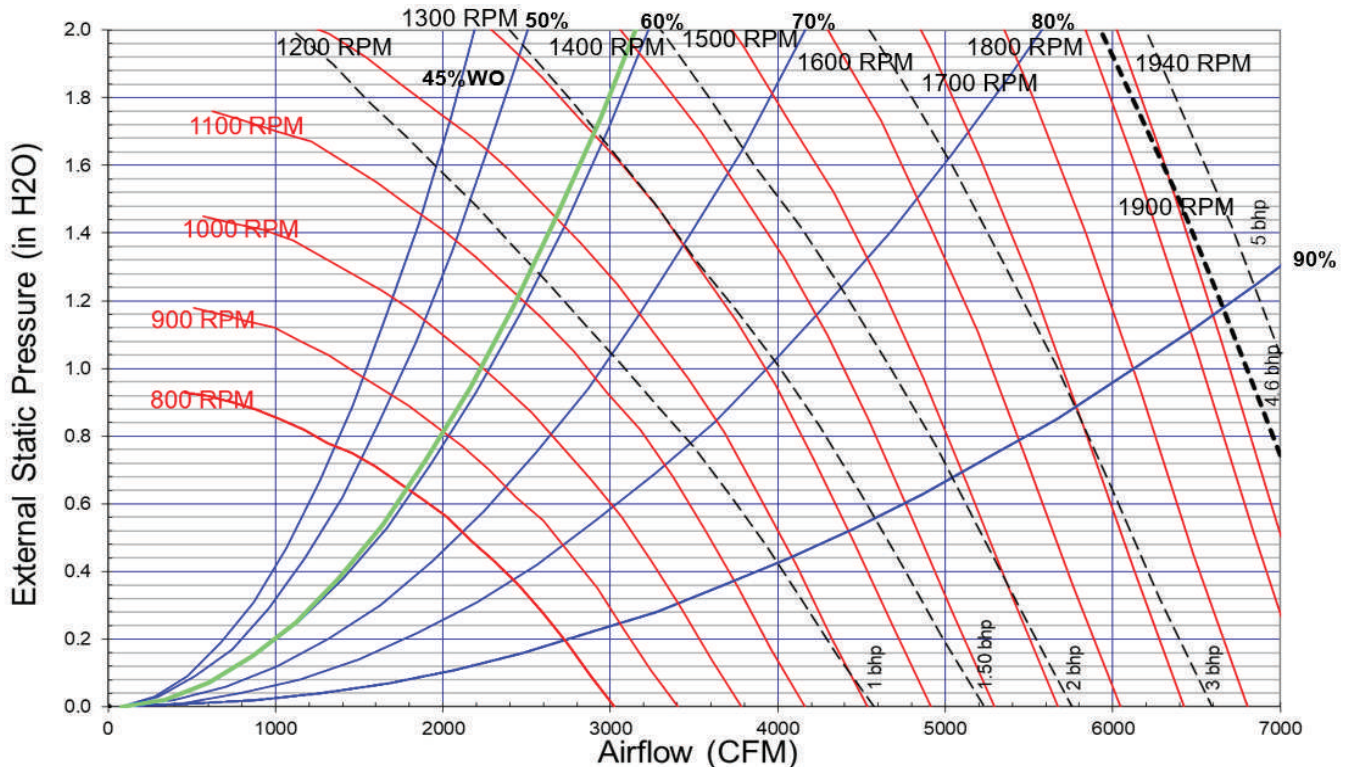
Available External Static Pressure (Inches of Water Gauge)										
CFM	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
5750	1462 1.72	1488 1.84	1510 1.95	1531 2.07	1551 2.18	1572 2.29	1591 2.41	1610 2.52	1629 2.64	1648 2.76
6000	1523 1.94	1548 2.07	1570 2.19	1590 2.31	1610 2.42	1630 2.54	1649 2.66	1667 2.78	1686 2.90	1704 3.02
Available External Static Pressure (Inches of Water Gauge)										
CFM	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
3750	1267 1.36	1293 1.45	1319 1.55	1345 1.64	1370 1.74	1394 1.85	1419 1.95	1443 2.06	1467 2.17	1490 2.28
4000	1314 1.5	1339 1.59	1363 1.69	1388 1.79	1412 1.9	1436 2	1459 2.11	1483 2.22	1506 2.33	1529 2.44
4250	1362 1.65	1386 1.75	1410 1.86	1433 1.96	1455 2.06	1479 2.17	1502 2.28	1524 2.39	1546 2.51	1568 2.62
4500	1411 1.82	1434 1.93	1457 2.03	1480 2.14	1502 2.25	1523 2.36	1545 2.47	1567 2.59	1589 2.7	1610 2.82
4750	1461 2.01	1484 2.12	1506 2.23	1527 2.34	1549 2.45	1570 2.56	1591 2.68	1612 2.8	1632 2.91	1653 3.04
5000	1512 2.21	1534 2.32	1556 2.43	1577 2.55	1597 2.67	1618 2.78	1638 2.9	1658 3.02	1678 3.15	1697 3.27
5250	1564 2.42	1585 2.54	1606 2.66	1627 2.78	1647 2.9	1667 3.02	1686 3.14	1706 3.27	1725 3.39	1744 3.52
5500	1616 2.65	1637 2.77	1657 2.9	1677 3.02	1697 3.15	1717 3.27	1736 3.4	1754 3.53	1773 3.66	1792 3.79
5750	1669 2.89	1689 3.02	1709 3.15	1729 3.28	1748 3.42	1767 3.55	1786 3.68	1804 3.81	1822 3.94	1840 4.08
6000	1722 3.15	1742 3.29	1762 3.42	1781 3.56	1799 3.7	1818 3.84	1836 3.97	1854 4.11	1872 4.25	1889 4.38

Notes:

1. Available External Static Pressure is the static pressure difference between the return duct and the supply duct plus the static pressure drop caused by accessories and options.
2. For direct drive evaporator fan speed (rpm), refer to the applicable table in the fan performance section.
3. Data includes pressure drop due to standard filters and wet coils. No accessories or options are included in pressure drop data.
4. To determine static pressure drop due to other options/accessories, refer to the applicable table in the fan performance section.
5. Direct drive fan motor heat is negligible.
6. Factory supplied motors, in commercial equipment, are definite purpose motors, specifically designed and tested to operate reliably and continuously at all cataloged conditions. Using the full horsepower range of our fan motors as shown in our tabular data will not result in nuisance tripping or premature motor failure. Our product's warranty will not be affected.

12.5 Ton Units — Horizontal

Figure 5. Fan curves — 12.5 ton, horizontal





Evaporator Fan Performance

Table 34. Evaporator fan performance - 12.5 ton (model THJ), horizontal

Available External Static Pressure (Inches of Water Gauge)										
CFM	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
3750	1026 0.63	1055 0.70	1083 0.77	1109 0.84	1135 0.92	1162 1.00	1188 1.08	1214 1.16	1239 1.25	1263 1.34
4000	1091 0.75	1118 0.83	1144 0.90	1170 0.98	1194 1.06	1219 1.14	1244 1.23	1268 1.31	1292 1.40	1316 1.49
4250	1155 0.89	1181 0.97	1206 1.05	1231 1.13	1254 1.21	1276 1.30	1300 1.39	1324 1.48	1346 1.57	1369 1.67
4500	1220 1.05	1245 1.13	1269 1.22	1292 1.30	1314 1.39	1336 1.48	1358 1.56	1380 1.66	1402 1.76	1424 1.86
4750	1284 1.22	1309 1.31	1332 1.40	1354 1.49	1375 1.58	1396 1.67	1417 1.76	1437 1.86	1459 1.96	1480 2.07
5000	1349 1.41	1373 1.51	1395 1.60	1416 1.70	1437 1.79	1457 1.89	1477 1.98	1496 2.08	1516 2.18	1536 2.29
5250	1414 1.62	1437 1.73	1458 1.83	1479 1.93	1499 2.02	1518 2.12	1537 2.23	1556 2.33	1574 2.43	1593 2.54
5500	1479 1.86	1502 1.97	1522 2.07	1542 2.17	1561 2.28	1580 2.38	1598 2.49	1616 2.59	1634 2.70	1651 2.81
5750	1544 2.11	1566 2.23	1586 2.34	1605 2.44	1623 2.55	1641 2.66	1659 2.77	1677 2.88	1694 2.99	1711 3.10
6000	1609 2.39	1631 2.51	1650 2.62	1668 2.74	1686 2.85	1704 2.96	1721 3.08	1738 3.19	1754 3.31	1771 3.42
Available External Static Pressure (Inches of Water Gauge)										
CFM	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
3750	1289 1.43	1316 1.53	1341 1.63	1366 1.73	1390 1.82	1413 1.92	1436 2.02	1457 2.11	1480 2.21	1503 2.32
4000	1339 1.59	1362 1.68	1387 1.78	1412 1.89	1436 1.99	1459 2.1	1481 2.2	1503 2.31	1524 2.41	1545 2.52
4250	1392 1.76	1414 1.86	1435 1.96	1458 2.06	1481 2.17	1504 2.29	1527 2.4	1549 2.51	1570 2.62	1590 2.73
4500	1445 1.96	1467 2.06	1488 2.16	1508 2.27	1528 2.37	1550 2.49	1573 2.6	1594 2.72	1615 2.84	1636 2.96
4750	1500 2.17	1520 2.28	1541 2.38	1561 2.49	1580 2.6	1600 2.71	1619 2.82	1640 2.94	1661 3.07	1682 3.19
5000	1556 2.4	1575 2.51	1594 2.62	1614 2.73	1633 2.85	1652 2.96	1671 3.08	1689 3.19	1707 3.31	1728 3.44
5250	1612 2.65	1631 2.76	1650 2.88	1668 2.99	1687 3.11	1705 3.23	1723 3.35	1741 3.47	1759 3.59	1776 3.71
5500	1669 2.92	1688 3.04	1706 3.16	1724 3.28	1741 3.4	1759 3.52	1776 3.65	1794 3.77	1811 3.9	1828 4.02
5750	1727 3.21	1745 3.33	1763 3.46	1780 3.58	1797 3.71	1814 3.83	1830 3.96	1847 4.09	1864 4.22	1881 4.35
6000	1787 3.54	1803 3.65	1820 3.78	1837 3.91	1853 4.04	1870 4.17	1886 4.3	1902 4.43	1918 4.57	- -

Notes:

1. Available External Static Pressure is the static pressure difference between the return duct and the supply duct plus the static pressure drop caused by accessories and options.
2. For direct drive evaporator fan speed (rpm), refer to the applicable table in the fan performance section.
3. Data includes pressure drop due to standard filters and wet coils. No accessories or options are included in pressure drop data.
4. To determine static pressure drop due to other options/accessories, refer to the applicable table in the fan performance section.
5. Direct drive fan motor heat is negligible.
6. Factory supplied motors, in commercial equipment, are definite purpose motors, specifically designed and tested to operate reliably and continuously at all cataloged conditions. Using the full horsepower range of our fan motors as shown in our tabular data will not result in nuisance tripping or premature motor failure. Our product's warranty will not be affected.

15 to 25 Ton Units — Downflow

Figure 6. Fan curves — 15 to 25 tons, downflow

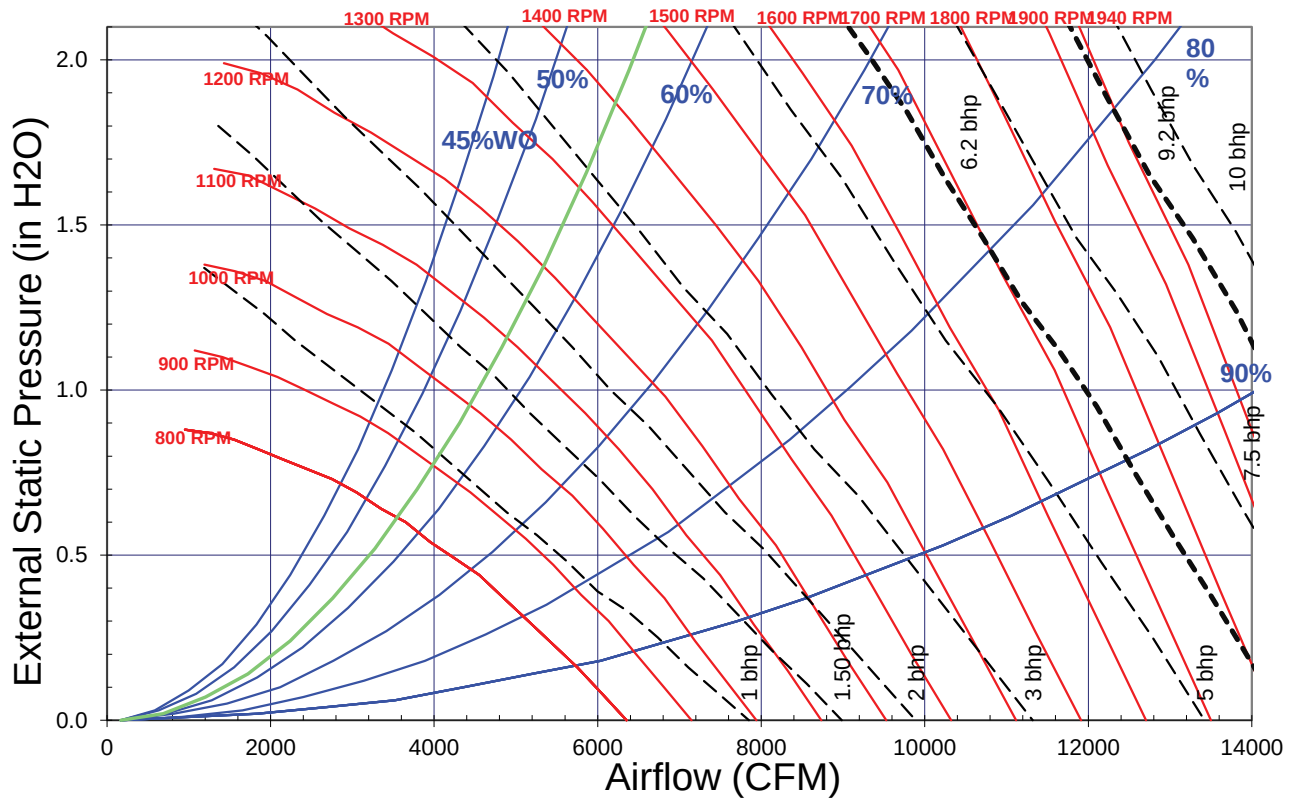


Table 35. Evaporator fan performance - 15 ton (model THJ), downflow

Available External Static Pressure (Inches of Water Gauge)											
CFM	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"	
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	
4500	628 0.29	684 0.40	734 0.51	779 0.62	825 0.74	869 0.86	910 0.98	948 1.11	984 1.24	1019 1.37	
4800	663 0.34	715 0.45	764 0.57	808 0.68	851 0.80	893 0.93	934 1.07	971 1.20	1007 1.33	1042 1.47	
5400	734 0.45	778 0.56	827 0.70	868 0.83	907 0.96	944 1.10	983 1.24	1020 1.39	1055 1.54	1089 1.69	
6000	804 0.58	847 0.71	888 0.85	930 1.01	967 1.15	1002 1.30	1036 1.44	1070 1.60	1104 1.76	1137 1.93	
6600	876 0.74	916 0.89	951 1.03	992 1.20	1028 1.37	1062 1.52	1094 1.68	1125 1.84	1155 2.01	1187 2.18	
7200	948 0.93	986 1.09	1020 1.25	1053 1.41	1090 1.61	1123 1.78	1154 1.96	1184 2.13	1213 2.30	1241 2.48	
Available External Static Pressure (Inches of Water Gauge)											
CFM	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"	
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	
4500	1054 1.51	1088 1.65	1120 1.79	1151 1.94	1182 2.09	1211 2.24	1239 2.39	1267 2.55	1294 2.71	1322 2.87	
4800	1075 1.61	1107 1.75	1140 1.9	1171 2.05	1201 2.21	1230 2.36	1258 2.52	1286 2.68	1313 2.85	1339 3.01	
5400	1121 1.84	1152 1.99	1182 2.14	1211 2.3	1240 2.46	1269 2.63	1297 2.8	1325 2.97	1351 3.15	1377 3.32	
6000	1169 2.09	1199 2.25	1229 2.42	1258 2.59	1285 2.76	1312 2.93	1338 3.11	1364 3.28	1391 3.47	1417 3.66	
6600	1218 2.37	1248 2.55	1277 2.73	1305 2.91	1332 3.09	1359 3.27	1385 3.46	1410 3.64	1435 3.83	1459 4.02	
7200	1269 2.66	1298 2.86	1326 3.05	1354 3.25	1381 3.45	1407 3.64	1432 3.84	1457 4.04	1482 4.24	1505 4.44	

Notes:

1. Available External Static Pressure is the static pressure difference between the return duct and the supply duct plus the static pressure drop caused by accessories and options.
2. For direct drive evaporator fan speed (rpm), refer to the applicable table in the fan performance section.
3. Data includes pressure drop due to standard filters and wet coils. No accessories or options are included in pressure drop data.
4. To determine static pressure drop due to other options/accessories, refer to the applicable table in the fan performance section.
5. Direct drive fan motor heat is negligible.
6. Factory supplied motors, in commercial equipment, are definite purpose motors, specifically designed and tested to operate reliably and continuously at all cataloged conditions. Using the full horsepower range of our fan motors as shown in our tabular data will not result in nuisance tripping or premature motor failure. Our product's warranty will not be affected.



Evaporator Fan Performance

Table 36. Evaporator fan performance - 17.5 ton (model THJ), downflow

Available External Static Pressure (Inches of Water Gauge)										
CFM	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
5250	716 0.42	761 0.53	811 0.67	853 0.79	893 0.92	931 1.05	970 1.20	1008 1.34	1043 1.48	1077 1.63
5600	757 0.49	801 0.61	847 0.75	888 0.89	927 1.02	963 1.16	1000 1.30	1036 1.46	1071 1.61	1105 1.76
6300	840 0.66	881 0.80	919 0.94	961 1.11	997 1.26	1032 1.41	1065 1.56	1096 1.71	1129 1.88	1162 2.05
7000	924 0.86	963 1.02	997 1.17	1033 1.34	1070 1.53	1103 1.69	1134 1.86	1164 2.03	1193 2.20	1221 2.37
7700	1009 1.11	1045 1.29	1077 1.46	1107 1.62	1141 1.82	1176 2.02	1205 2.21	1234 2.39	1262 2.58	1289 2.76
8400	1094 1.41	1127 1.60	1159 1.79	1187 1.97	1214 2.15	1247 2.37	1278 2.60	1306 2.80	1333 3.00	1358 3.20
Available External Static Pressure (Inches of Water Gauge)										
CFM	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
5250	1109 1.78	1141 1.93	1171 2.08	1201 2.24	1230 2.4	1259 2.56	1288 2.73	1315 2.9	1342 3.07	1368 3.24
5600	1137 1.92	1168 2.07	1198 2.23	1227 2.39	1255 2.56	1283 2.73	1311 2.9	1338 3.07	1364 3.25	1390 3.43
6300	1193 2.23	1223 2.4	1253 2.57	1281 2.74	1309 2.92	1336 3.1	1362 3.28	1387 3.46	1412 3.64	1437 3.83
7000	1252 2.56	1281 2.75	1310 2.94	1337 3.14	1364 3.33	1391 3.52	1416 3.71	1441 3.9	1466 4.1	1490 4.3
7700	1315 2.95	1341 3.14	1368 3.34	1395 3.55	1422 3.76	1447 3.98	1473 4.19	1497 4.4	1521 4.6	1545 4.82
8400	1384 3.4	1409 3.6	1433 3.81	1456 4.01	1481 4.23	1506 4.46	1531 4.69	1555 4.92	1578 5.15	1601 5.38

Notes:

1. Available External Static Pressure is the static pressure difference between the return duct and the supply duct plus the static pressure drop caused by accessories and options.
2. For direct drive evaporator fan speed (rpm), refer to the applicable table in the fan performance section.
3. Data includes pressure drop due to standard filters and wet coils. No accessories or options are included in pressure drop data.
4. To determine static pressure drop due to other options/accessories, refer to the applicable table in the fan performance section.
5. Direct drive fan motor heat is negligible.
6. Factory supplied motors, in commercial equipment, are definite purpose motors, specifically designed and tested to operate reliably and continuously at all cataloged conditions. Using the full horsepower range of our fan motors as shown in our tabular data will not result in nuisance tripping or premature motor failure. Our product's warranty will not be affected.

Table 37. Evaporator fan performance - 20 ton (model THJ), downflow

Available External Static Pressure (Inches of Water Gauge)										
CFM	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
6000	804 0.58	847 0.71	888 0.85	930 1.01	967 1.15	1002 1.30	1036 1.44	1070 1.60	1104 1.76	1137 1.93
6400	852 0.68	893 0.83	929 0.97	971 1.14	1008 1.29	1042 1.45	1074 1.60	1106 1.76	1138 1.92	1170 2.10
7200	948 0.93	986 1.09	1020 1.25	1053 1.41	1090 1.61	1123 1.78	1154 1.96	1184 2.13	1213 2.30	1241 2.48
8000	1045 1.23	1080 1.41	1112 1.59	1141 1.76	1172 1.95	1206 2.17	1237 2.37	1265 2.56	1292 2.75	1319 2.94
8800	1143 1.60	1175 1.80	1206 2.00	1233 2.19	1260 2.38	1288 2.59	1319 2.82	1348 3.06	1374 3.26	1399 3.47
9600	1241 2.04	1271 2.25	1300 2.47	1327 2.69	1351 2.89	1375 3.10	1401 3.32	1429 3.58	1457 3.84	1482 4.08
Available External Static Pressure (Inches of Water Gauge)										
CFM	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
6000	1169 2.09	1199 2.25	1229 2.42	1258 2.59	1285 2.76	1312 2.93	1338 3.11	1364 3.28	1391 3.47	1417 3.66
6400	1201 2.27	1232 2.45	1261 2.62	1289 2.8	1317 2.98	1343 3.15	1369 3.34	1395 3.52	1419 3.71	1444 3.89
7200	1269 2.66	1298 2.86	1326 3.05	1354 3.25	1381 3.45	1407 3.64	1432 3.84	1457 4.04	1482 4.24	1505 4.44
8000	1345 3.14	1370 3.33	1394 3.53	1421 3.74	1447 3.96	1472 4.18	1497 4.4	1521 4.62	1545 4.84	1569 5.05
8800	1424 3.69	1448 3.9	1472 4.11	1495 4.32	1518 4.54	1540 4.75	1564 4.99	1588 5.23	1611 5.47	1634 5.72
9600	1506 4.3	1529 4.53	1551 4.76	1573 4.99	1595 5.22	1617 5.46	1638 5.69	1659 5.92	-	-

Notes:

1. Available External Static Pressure is the static pressure difference between the return duct and the supply duct plus the static pressure drop caused by accessories and options.
2. For direct drive evaporator fan speed (rpm), refer to the applicable table in the fan performance section.
3. Data includes pressure drop due to standard filters and wet coils. No accessories or options are included in pressure drop data.
4. To determine static pressure drop due to other options/accessories, refer to the applicable table in the fan performance section.
5. Direct drive fan motor heat is negligible.
6. Factory supplied motors, in commercial equipment, are definite purpose motors, specifically designed and tested to operate reliably and continuously at all cataloged conditions. Using the full horsepower range of our fan motors as shown in our tabular data will not result in nuisance tripping or premature motor failure. Our product's warranty will not be affected.

Table 38. Evaporator fan performance - 25 ton (model THJ), downflow

Available External Static Pressure (Inches of Water Gauge)										
CFM	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
3-hp Standard Motor										
7500	984 1.03	1021 1.21	1054 1.37	1085 1.53	1121 1.73	1155 1.92	1185 2.10	1214 2.28	1242 2.46	1270 2.64
8000	1045 1.23	1080 1.41	1112 1.59	1141 1.76	1172 1.95	1206 2.17	1237 2.37	1265 2.56	1292 2.75	1319 2.94
9000	1167 1.70	1199 1.90	1229 2.11	1257 2.31	1283 2.50	1309 2.70	1339 2.94	1368 3.19	1395 3.40	1420 3.62
10000	1290 2.28	1319 2.50	1347 2.73	1374 2.96	1398 3.18	1421 3.39	1444 3.61	1471 3.86	1498 4.13	1524 4.40
11000	1413 2.99	1440 3.23	1466 3.47	1491 3.73	1515 3.98	1536 4.22	1558 4.45	1579 4.69	1601 4.94	1626 5.24
12000	1537 3.83	1561 4.09	1585 4.36	1609 4.63	1631 4.91	1653 5.18	1673 5.44	1693 5.69	1712 5.95	- -
Available External Static Pressure (Inches of Water Gauge)										
CFM	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
3-hp Standard Motor										
7500	1296 2.83	1323 3.02	1351 3.22	1379 3.43	1405 3.64	1431 3.84	1456 4.05	1481 4.25	1505 4.46	1529 4.66
8000	1345 3.14	1370 3.33	1394 3.53	1421 3.74	1447 3.96	1472 4.18	1497 4.4	1521 4.62	1545 4.84	1569 5.05
9000	1444 3.83	1468 4.05	1491 4.26	1514 4.48	1537 4.7	1559 4.92	1581 5.15	1605 5.39	1628 5.64	1651 5.89
10000	1547 4.64	1570 4.88	1592 5.12	1614 5.36	1635 5.6	1656 5.84	- -	- -	- -	- -
11000	1650 5.53	1674 5.83	- -	- -	- -	- -	- -	- -	- -	- -
12000	- -	- -	- -	- -	- -	- -	- -	- -	- -	- -

Notes:

1. Available External Static Pressure is the static pressure difference between the return duct and the supply duct plus the static pressure drop caused by accessories and options.
2. For direct drive evaporator fan speed (rpm), refer to the applicable table in the fan performance section.
3. Data includes pressure drop due to standard filters and wet coils. No accessories or options are included in pressure drop data.
4. To determine static pressure drop due to other options/accessories, refer to the applicable table in the fan performance section.
5. Direct drive fan motor heat is negligible.
6. Factory supplied motors, in commercial equipment, are definite purpose motors, specifically designed and tested to operate reliably and continuously at all cataloged conditions. Using the full horsepower range of our fan motors as shown in our tabular data will not result in nuisance tripping or premature motor failure. Our product's warranty will not be affected.

Table 39. Evaporator fan performance - 25 ton (model THJ), downflow, oversized/high static motor option

Available External Static Pressure (Inches of Water Gauge)										
CFM	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
5-hp Oversized Motor										
7500	1114 1.60	1146 1.77	1176 1.94	1207 2.12	1237 2.30	1265 2.48	1294 2.67	1322 2.86	1350 3.05	1376 3.24
8000	1184 1.91	1214 2.09	1243 2.28	1272 2.46	1300 2.66	1328 2.85	1354 3.04	1382 3.25	1408 3.45	1434 3.65
9000	1324 2.67	1352 2.87	1378 3.08	1403 3.28	1429 3.49	1454 3.71	1479 3.93	1503 4.15	1527 4.37	1551 4.59
10000	1466 3.61	1490 3.84	1514 4.06	1538 4.29	1560 4.52	1584 4.75	1607 4.99	1629 5.23	1651 5.47	1673 5.72
11000	1607 4.76	1630 5.00	1652 5.25	1674 5.50	1695 5.75	1715 6.00	1736 6.26	1758 6.52	1778 6.78	1799 7.05
12000	1749 6.13	1770 6.39	1791 6.66	1811 6.93	1831 7.20	1850 7.48	1869 7.75	1888 8.03	1907 8.31	1927 8.6
Available External Static Pressure (Inches of Water Gauge)										
CFM	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
5-hp Oversized Motor										
7500	1401 3.44	1426 3.63	1450 3.83	1474 4.03	1496 4.23	1519 4.43	1543 4.65	1568 4.87	1592 5.09	1616 5.31
8000	1458 3.86	1483 4.06	1506 4.27	1529 4.48	1551 4.69	1573 4.91	1595 5.12	1616 5.34	1638 5.56	1662 5.8
9000	1575 4.82	1598 5.05	1620 5.28	1642 5.51	1664 5.74	1685 5.97	1706 6.21	1726 6.45	1746 6.69	1765 6.92
10000	1694 5.96	1716 6.21	1738 6.46	1759 6.72	1779 6.97	1799 7.23	1819 7.48	1839 7.74	1858 8	1877 8.26
11000	1819 7.31	1838 7.58	1858 7.85	1878 8.13	1897 8.4	1917 8.68	1936 8.96	- -	- -	- -



Evaporator Fan Performance

Table 39. Evaporator fan performance - 25 ton (model THJ), downflow, oversized/high static motor option (continued)

Available External Static Pressure (Inches of Water Gauge)										
CFM	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
5-hp Oversized Motor										
12000	- -	- -	- -	- -	- -	- -	- -	- -	- -	- -

Notes:

1. Available External Static Pressure is the static pressure difference between the return duct and the supply duct plus the static pressure drop caused by accessories and options.
2. For direct drive evaporator fan speed (rpm), refer to the applicable table in the fan performance section.
3. Data includes pressure drop due to standard filters and wet coils. No accessories or options are included in pressure drop data.
4. To determine static pressure drop due to other options/accessories, refer to the applicable table in the fan performance section.
5. Direct drive fan motor heat is negligible.
6. Factory supplied motors, in commercial equipment, are definite purpose motors, specifically designed and tested to operate reliably and continuously at all cataloged conditions. Using the full horsepower range of our fan motors as shown in our tabular data will not result in nuisance tripping or premature motor failure. Our product's warranty will not be affected.

15 to 25 Ton Units — Horizontal

Figure 7. Fan curves — 15 to 25 tons, horizontal

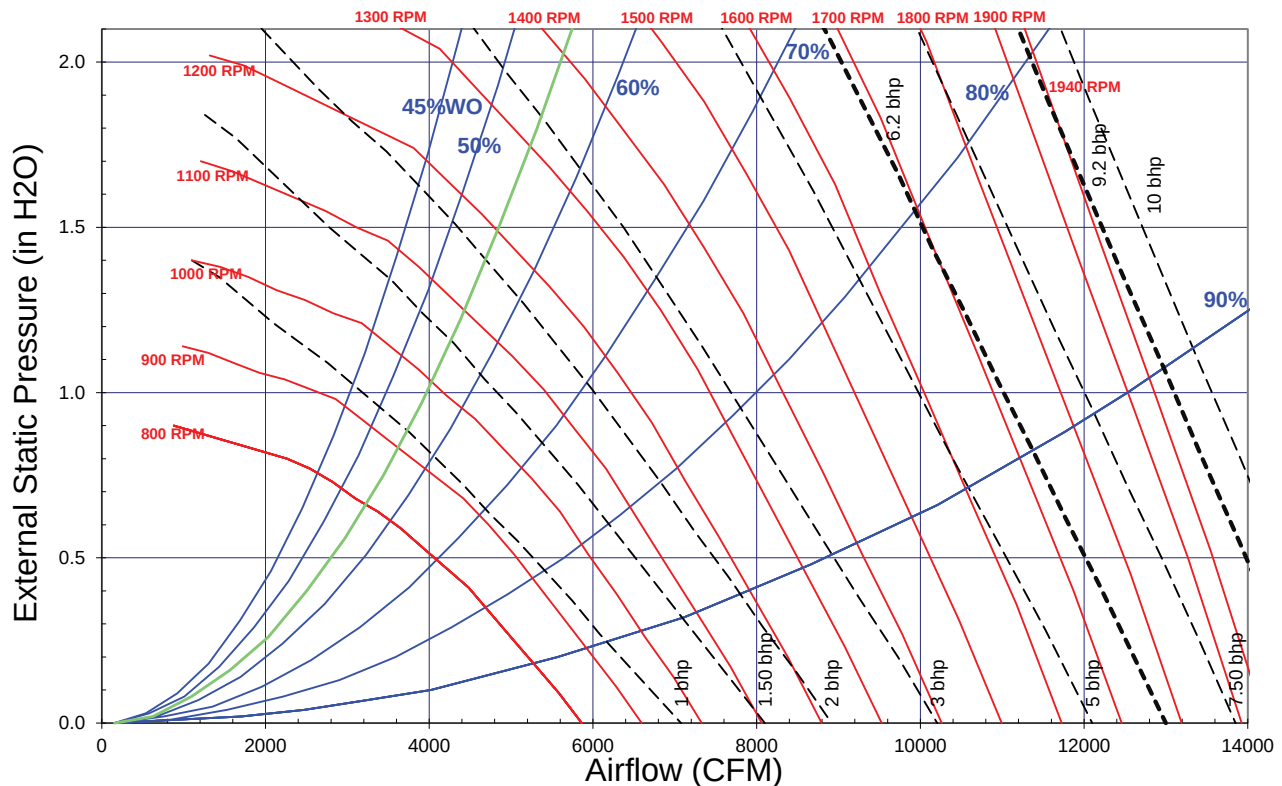


Table 40. Evaporator fan performance - 15 ton (model THJ), horizontal

Available External Static Pressure (Inches of Water Gauge)										
CFM	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
4500	667 0.36	716 0.46	760 0.56	800 0.66	841 0.77	880 0.88	917 1.00	954 1.12	990 1.25	1024 1.38
4800	705 0.42	752 0.52	795 0.63	833 0.74	870 0.85	908 0.97	944 1.09	979 1.22	1014 1.35	1048 1.48
5400	781 0.56	825 0.68	864 0.80	901 0.92	935 1.05	968 1.17	1002 1.31	1035 1.44	1067 1.58	1098 1.72
6000	858 0.74	899 0.87	936 1.01	971 1.14	1003 1.28	1034 1.41	1063 1.55	1094 1.69	1124 1.84	1153 1.99
6600	936 0.95	973 1.10	1008 1.25	1041 1.39	1073 1.54	1102 1.69	1130 1.84	1157 1.99	1184 2.15	1212 2.31
7200	1015 1.21	1049 1.36	1082 1.53	1113 1.69	1143 1.85	1171 2.01	1199 2.17	1225 2.34	1250 2.50	1274 2.66

Table 40. Evaporator fan performance - 15 ton (model THJ), horizontal (continued)

Available External Static Pressure (Inches of Water Gauge)										
	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
CFM	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
4500	1057 1.51	1089 1.64	1120 1.78	1150 1.92	1180 2.06	1208 2.21	1236 2.36	1263 2.51	1289 2.66	1315 2.82
4800	1081 1.62	1112 1.76	1143 1.9	1172 2.04	1200 2.19	1229 2.34	1256 2.5	1283 2.65	1310 2.81	1335 2.97
5400	1128 1.86	1159 2.01	1189 2.16	1218 2.32	1246 2.47	1273 2.63	1300 2.79	1325 2.95	1351 3.12	1376 3.29
6000	1182 2.15	1210 2.3	1237 2.46	1265 2.62	1293 2.79	1320 2.96	1346 3.13	1372 3.3	1396 3.48	1421 3.65
6600	1240 2.47	1266 2.64	1293 2.81	1318 2.97	1343 3.15	1368 3.32	1393 3.5	1419 3.69	1443 3.87	1467 4.06
7200	1299 2.84	1325 3.01	1350 3.19	1374 3.37	1399 3.56	1423 3.74	1446 3.93	1469 4.11	1491 4.3	1515 4.5

Notes:

1. Available External Static Pressure is the static pressure difference between the return duct and the supply duct plus the static pressure drop caused by accessories and options.
2. For direct drive evaporator fan speed (rpm), refer to the applicable table in the fan performance section.
3. Data includes pressure drop due to standard filters and wet coils. No accessories or options are included in pressure drop data.
4. To determine static pressure drop due to other options/accessories, refer to the applicable table in the fan performance section.
5. Direct drive fan motor heat is negligible.
6. Factory supplied motors, in commercial equipment, are definite purpose motors, specifically designed and tested to operate reliably and continuously at all cataloged conditions. Using the full horsepower range of our fan motors as shown in our tabular data will not result in nuisance tripping or premature motor failure. Our product's warranty will not be affected.

Table 41. Evaporator fan performance - 17.5 ton (model THJ), horizontal

Available External Static Pressure (Inches of Water Gauge)										
	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
CFM	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
5250	762 0.52	806 0.64	847 0.76	884 0.88	918 0.99	953 1.12	987 1.25	1021 1.38	1053 1.52	1084 1.65
5600	807 0.62	849 0.74	888 0.87	924 0.99	958 1.12	989 1.25	1022 1.38	1054 1.52	1086 1.66	1116 1.81
6300	897 0.84	936 0.98	972 1.12	1006 1.26	1038 1.40	1068 1.55	1097 1.69	1124 1.84	1154 1.99	1183 2.15
7000	989 1.12	1024 1.27	1057 1.43	1089 1.58	1119 1.74	1148 1.90	1176 2.06	1202 2.22	1227 2.38	1252 2.54
7700	1081 1.45	1113 1.62	1144 1.79	1174 1.96	1202 2.14	1230 2.31	1256 2.48	1282 2.66	1306 2.83	1330 3.01
8400	1175 1.85	1203 2.03	1232 2.22	1260 2.41	1287 2.59	1313 2.78	1338 2.97	1363 3.16	1386 3.35	1409 3.54
Available External Static Pressure (Inches of Water Gauge)										
	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
CFM	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
5250	1116 1.8	1147 1.95	1177 2.09	1206 2.25	1234 2.4	1262 2.56	1288 2.71	1314 2.87	1340 3.04	1366 3.21
5600	1145 1.95	1175 2.1	1205 2.26	1234 2.42	1262 2.58	1289 2.74	1315 2.9	1341 3.07	1366 3.23	1390 3.4
6300	1210 2.31	1238 2.47	1265 2.63	1291 2.79	1317 2.96	1344 3.13	1370 3.31	1395 3.49	1420 3.67	1444 3.85
7000	1279 2.71	1305 2.88	1331 3.06	1355 3.24	1380 3.41	1404 3.59	1427 3.78	1451 3.96	1475 4.15	1499 4.35
7700	1353 3.18	1375 3.36	1400 3.55	1424 3.74	1447 3.93	1470 4.12	1492 4.32	1515 4.52	1537 4.71	1558 4.91
8400	1431 3.73	1453 3.92	1474 4.11	1494 4.31	1516 4.51	1539 4.72	1560 4.93	1582 5.13	1602 5.35	1623 5.56

Notes:

1. Available External Static Pressure is the static pressure difference between the return duct and the supply duct plus the static pressure drop caused by accessories and options.
2. For direct drive evaporator fan speed (rpm), refer to the applicable table in the fan performance section.
3. Data includes pressure drop due to standard filters and wet coils. No accessories or options are included in pressure drop data.
4. To determine static pressure drop due to other options/accessories, refer to the applicable table in the fan performance section.
5. Direct drive fan motor heat is negligible.
6. Factory supplied motors, in commercial equipment, are definite purpose motors, specifically designed and tested to operate reliably and continuously at all cataloged conditions. Using the full horsepower range of our fan motors as shown in our tabular data will not result in nuisance tripping or premature motor failure. Our product's warranty will not be affected.

Table 42. Evaporator fan performance - 20 ton (model THJ), horizontal

Available External Static Pressure (Inches of Water Gauge)										
	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
CFM	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
6000	858 0.74	899 0.87	936 1.01	971 1.14	1003 1.28	1034 1.41	1063 1.55	1094 1.69	1124 1.84	1153 1.99
6400	910 0.88	948 1.02	984 1.16	1018 1.30	1049 1.45	1079 1.59	1108 1.74	1135 1.89	1164 2.04	1192 2.20
7200	1015 1.21	1049 1.36	1082 1.53	1113 1.69	1143 1.85	1171 2.01	1199 2.17	1225 2.34	1250 2.50	1274 2.66
8000	1121 1.61	1151 1.79	1182 1.97	1211 2.14	1239 2.32	1265 2.50	1291 2.68	1316 2.86	1341 3.05	1364 3.23
8800	1228 2.11	1254 2.30	1283 2.49	1310 2.69	1336 2.89	1361 3.08	1386 3.28	1409 3.48	1433 3.68	1455 3.88
9600	1335 2.70	1359 2.90	1385 3.12	1410 3.33	1435 3.54	1459 3.76	1482 3.97	1504 4.19	1526 4.40	1548 4.62



Evaporator Fan Performance

Table 42. Evaporator fan performance - 20 ton (model THJ), horizontal (continued)

Available External Static Pressure (Inches of Water Gauge)										
CFM	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
6000	1182 2.15	1210 2.3	1237 2.46	1265 2.62	1293 2.79	1320 2.96	1346 3.13	1372 3.3	1396 3.48	1421 3.65
6400	1220 2.36	1247 2.52	1274 2.69	1300 2.85	1325 3.02	1352 3.2	1378 3.37	1403 3.55	1427 3.74	1452 3.92
7200	1299 2.84	1325 3.01	1350 3.19	1374 3.37	1399 3.56	1423 3.74	1446 3.93	1469 4.11	1491 4.3	1515 4.5
8000	1386 3.41	1408 3.59	1430 3.78	1453 3.97	1476 4.17	1499 4.37	1521 4.57	1543 4.77	1565 4.98	1586 5.18
8800	1477 4.08	1498 4.27	1518 4.47	1539 4.67	1558 4.88	1579 5.09	1600 5.3	1621 5.52	1642 5.74	1662 5.96
9600	1569 4.84	1589 5.05	1609 5.27	1629 5.49	1648 5.71	1666 5.92	- -	- -	- -	- -

Notes:

1. Available External Static Pressure is the static pressure difference between the return duct and the supply duct plus the static pressure drop caused by accessories and options.
2. For direct drive evaporator fan speed (rpm), refer to the applicable table in the fan performance section.
3. Data includes pressure drop due to standard filters and wet coils. No accessories or options are included in pressure drop data.
4. To determine static pressure drop due to other options/accessories, refer to the applicable table in the fan performance section.
5. Direct drive fan motor heat is negligible.
6. Factory supplied motors, in commercial equipment, are definite purpose motors, specifically designed and tested to operate reliably and continuously at all cataloged conditions. Using the full horsepower range of our fan motors as shown in our tabular data will not result in nuisance tripping or premature motor failure. Our product's warranty will not be affected.

Table 43. Evaporator fan performance - 25 ton (model THJ), horizontal

Available External Static Pressure (Inches of Water Gauge)										
CFM	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
3-hp Standard Motor										
7500	1055 1.35	1087 1.51	1119 1.68	1150 1.85	1178 2.02	1206 2.18	1233 2.35	1259 2.52	1284 2.69	1307 2.86
8000	1121 1.61	1151 1.79	1182 1.97	1211 2.14	1239 2.32	1265 2.50	1291 2.68	1316 2.86	1341 3.05	1364 3.23
9000	1255 2.25	1280 2.44	1308 2.64	1335 2.84	1361 3.04	1385 3.24	1409 3.45	1433 3.65	1456 3.85	1478 4.05
10000	1388 3.03	1412 3.25	1436 3.46	1461 3.69	1485 3.91	1508 4.13	1530 4.36	1552 4.58	1574 4.81	1595 5.03
11000	1523 3.99	1544 4.22	1565 4.46	1588 4.70	1610 4.95	1632 5.19	1653 5.44	1674 5.68	1694 5.93	- -
12000	1657 5.13	1677 5.38	1696 5.64	1716 5.90	- -	- -	- -	- -	- -	- -
Available External Static Pressure (Inches of Water Gauge)										
CFM	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
3-hp Standard Motor										
7500	1330 3.04	1355 3.22	1380 3.4	1404 3.59	1427 3.78	1451 3.97	1474 4.16	1496 4.35	1518 4.54	1540 4.74
8000	1386 3.41	1408 3.59	1430 3.78	1453 3.97	1476 4.17	1499 4.37	1521 4.57	1543 4.77	1565 4.98	1586 5.18
9000	1500 4.26	1521 4.46	1541 4.66	1561 4.87	1580 5.07	1599 5.28	1620 5.5	1641 5.72	1661 5.94	- -
10000	1615 5.25	1635 5.48	1655 5.71	1675 5.94	- -	- -	- -	- -	- -	- -
11000	- -	- -	- -	- -	- -	- -	- -	- -	- -	- -
12000	- -	- -	- -	- -	- -	- -	- -	- -	- -	- -

Notes:

1. Available External Static Pressure is the static pressure difference between the return duct and the supply duct plus the static pressure drop caused by accessories and options.
2. For direct drive evaporator fan speed (rpm), refer to the applicable table in the fan performance section.
3. Data includes pressure drop due to standard filters and wet coils. No accessories or options are included in pressure drop data.
4. To determine static pressure drop due to other options/accessories, refer to the applicable table in the fan performance section.
5. Direct drive fan motor heat is negligible.
6. Factory supplied motors, in commercial equipment, are definite purpose motors, specifically designed and tested to operate reliably and continuously at all cataloged conditions. Using the full horsepower range of our fan motors as shown in our tabular data will not result in nuisance tripping or premature motor failure. Our product's warranty will not be affected.

Table 44. Evaporator fan performance - 25 ton (model THJ), horizontal, oversized/high static motor option

Available External Static Pressure (Inches of Water Gauge)										
CFM	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
5-hp Oversized Motor										
7500	1114 1.60	1146 1.77	1176 1.94	1207 2.12	1237 2.30	1265 2.48	1294 2.67	1322 2.86	1350 3.05	1376 3.24
8000	1184 1.91	1214 2.09	1243 2.28	1272 2.46	1300 2.66	1328 2.85	1354 3.04	1382 3.25	1408 3.45	1434 3.65
9000	1324 2.67	1352 2.87	1378 3.08	1403 3.28	1429 3.49	1454 3.71	1479 3.93	1503 4.15	1527 4.37	1551 4.59
10000	1466 3.61	1490 3.84	1514 4.06	1538 4.29	1560 4.52	1584 4.75	1607 4.99	1629 5.23	1651 5.47	1673 5.72

Table 44. Evaporator fan performance - 25 ton (model THJ), horizontal, oversized/high static motor option (continued)

Available External Static Pressure (Inches of Water Gauge)										
CFM	0.10"	0.20"	0.30"	0.40"	0.50"	0.60"	0.70"	0.80"	0.90"	1.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
5-hp Oversized Motor										
11000	1607 4.76	1630 5.00	1652 5.25	1674 5.50	1695 5.75	1715 6.00	1736 6.26	1758 6.52	1778 6.78	1799 7.05
12000	1749 6.13	1770 6.39	1791 6.66	1811 6.93	1831 7.20	1850 7.48	1869 7.75	1888 8.03	1907 8.31	1927 8.6
Available External Static Pressure (Inches of Water Gauge)										
CFM	1.10"	1.20"	1.30"	1.40"	1.50"	1.60"	1.70"	1.80"	1.90"	2.00"
	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP	RPM BHP
5-hp Oversized Motor										
7500	1401 3.44	1426 3.63	1450 3.83	1474 4.03	1496 4.23	1519 4.43	1543 4.65	1568 4.87	1592 5.09	1616 5.31
8000	1458 3.86	1483 4.06	1506 4.27	1529 4.48	1551 4.69	1573 4.91	1595 5.12	1616 5.34	1638 5.56	1662 5.8
9000	1575 4.82	1598 5.05	1620 5.28	1642 5.51	1664 5.74	1685 5.97	1706 6.21	1726 6.45	1746 6.69	1765 6.92
10000	1694 5.96	1716 6.21	1738 6.46	1759 6.72	1779 6.97	1799 7.23	1819 7.48	1839 7.74	1858 8	1877 8.26
11000	1819 7.31	1838 7.58	1858 7.85	1878 8.13	1897 8.4	1917 8.68	1936 8.96	- -	- -	- -
12000	- -	- -	- -	- -	- -	- -	- -	- -	- -	- -

Notes:

1. Available External Static Pressure is the static pressure difference between the return duct and the supply duct plus the static pressure drop caused by accessories and options.
2. For direct drive evaporator fan speed (rpm), refer to the applicable table in the fan performance section.
3. Data includes pressure drop due to standard filters and wet coils. No accessories or options are included in pressure drop data.
4. To determine static pressure drop due to other options/accessories, refer to the applicable table in the fan performance section.
5. Direct drive fan motor heat is negligible.
6. Factory supplied motors, in commercial equipment, are definite purpose motors, specifically designed and tested to operate reliably and continuously at all cataloged conditions. Using the full horsepower range of our fan motors as shown in our tabular data will not result in nuisance tripping or premature motor failure. Our product's warranty will not be affected.



Fan Performance

Table 45. Outdoor sound power level

Tons	Unit Model Number	Octave Center Frequency								Overall dBA
		63	125	250	500	1000	2000	4000	8000	
6	THJ072A*	83	84	84	85	82	76	73	67	86
7.5	THJ090A*	83	84	84	85	82	76	73	67	86
8.5	THJ102A*	83	84	84	85	82	76	73	67	86
10	THJ120A*	87	88	86	83	81	77	73	68	86
12.5	THJ150A*	89	92	89	87	84	80	75	68	89
15	THJ180A*	83	87	87	85	82	77	74	69	87
17.5	THJ210A*	89	89	91	89	86	82	79	73	91
20	THJ240A*	89	89	91	89	86	82	79	73	91
25	THJ300A*	94	90	92	91	88	84	81	75	93

Notes:

1. Outdoor sound rating shown is tested in accordance with AHRI 270/370-2015. For additional information reference the outdoor sound power level data in the performance section.
2. Taken in accordance with AHRI 270/370-2015.
3. Indoor sound in accordance with AHRI 260 is available through Trane's selection software.

Table 46. Static pressure drop through accessories (inches water column) - high efficiency

Tons	Unit Model Number	cfm	Standard Filters ^(a)	2" MERV 8 Filter ^(b)	2" MERV 13 Filter ^(b)	Reheat Coil	Economizer with OA/RA Dampers ^(c)								Electric Heater			
							Downflow		Horizontal		Low Leak Downflow ^(d)		Low Leak Horizontal		Accessory (kW)			
							100% OA	100% RA	100% OA	100% RA	100% OA	100% RA	100% OA	100% RA	9-18	27-36	54	72
6	THJ072A	1800	0.03	0.04	0.07	0.04	0.05	0.01	0.04	0.02	0.09	0.00	-	-	0.01	0.02	N/A	N/A
		2400	0.04	0.06	0.10	0.06	0.10	0.01	0.06	0.03	0.16	0.01	-	-	0.02	0.03	N/A	N/A
		2880	0.04	0.07	0.13	0.07	0.14	0.02	0.08	0.04	0.24	0.01	-	-	0.03	0.03	N/A	N/A
7.5	THJ090A	2250	0.03	0.05	0.09	0.05	0.09	0.01	0.05	0.02	0.14	0.01	-	-	0.02	0.02	N/A	N/A
		3000	0.05	0.08	0.13	0.07	0.15	0.02	0.09	0.04	0.26	0.01	-	-	0.03	0.03	N/A	N/A
		3600	0.05	0.10	0.17	0.09	0.22	0.02	0.12	0.06	0.39	0.02	-	-	0.04	0.05	N/A	N/A
8.5	THJ102A	2550	0.04	0.06	0.11	0.06	0.11	0.01	0.06	0.03	0.19	0.01	-	-	0.02	0.03	N/A	N/A
		3400	0.05	0.09	0.16	0.09	0.20	0.02	0.11	0.05	0.34	0.02	-	-	0.03	0.04	N/A	N/A
		4080	0.06	0.12	0.20	0.1	0.28	0.03	0.15	0.07	0.50	0.03	-	-	0.05	0.06	N/A	N/A
10	THJ120A	3000	0.05	0.08	0.13	0.07	0.15	0.02	0.09	0.04	0.26	0.01	-	-	0.02	0.03	0.05	N/A
		4000	0.06	0.12	0.20	0.10	0.27	0.03	0.15	0.07	0.48	0.03	-	-	0.02	0.03	0.05	N/A
		4800	0.07	0.16	0.25	0.12	0.39	0.03	0.20	0.09	0.71	0.05	-	-	0.03	0.04	0.06	N/A
12.5	THJ150A	3750	0.05	0.09	0.16	0.08	0.24	0.02	0.13	0.06	0.42	0.02	-	-	0.02	0.03	0.05	N/A
		5000	0.07	0.13	0.22	0.1	0.42	0.04	0.22	0.10	0.78	0.06	-	-	0.02	0.03	0.05	N/A
		6000	0.09	0.16	0.27	0.12	0.60	0.05	0.31	0.13	1.16	0.10	-	-	0.03	0.04	0.06	N/A
15	THJ180A	4500	0.03	0.05	0.09	0.04	0.13	0.02	0.13	0.02	0.14	0.11	0.07	0.12	0.01	0.02	0.02	N/A
		6000	0.04	0.08	0.13	0.05	0.20	0.04	0.20	0.04	0.23	0.17	0.12	0.18	0.01	0.04	0.04	N/A
		7200	0.06	0.10	0.17	0.07	0.27	0.05	0.27	0.05	0.32	0.23	0.16	0.25	0.02	0.06	0.06	N/A
17.5	THJ210A	5250	0.04	0.06	0.11	0.05	0.16	0.03	0.16	0.03	0.19	0.14	0.09	0.15	N/A	0.03	0.03	0.03
		7000	0.05	0.10	0.17	0.06	0.26	0.05	0.26	0.05	0.30	0.22	0.15	0.24	N/A	0.06	0.06	0.06
		8400	0.07	0.13	0.22	0.08	0.35	0.06	0.35	0.06	0.42	0.29	0.21	0.33	N/A	0.09	0.09	0.09
20	THJ240A	6000	0.04	0.08	0.13	0.05	0.20	0.04	0.20	0.04	0.23	0.17	0.12	0.18	N/A	0.04	0.04	0.04
		8000	0.07	0.12	0.21	0.08	0.32	0.06	0.32	0.06	0.39	0.27	0.19	0.3	N/A	0.08	0.08	0.08
		9600	0.09	0.16	0.27	0.10	0.44	0.07	0.44	0.07	0.54	0.37	0.27	0.41	N/A	0.12	0.12	0.12
25	THJ300A	7500	0.06	0.11	0.19	0.07	0.29	0.05	0.29	0.05	0.34	0.24	0.17	0.27	N/A	0.07	0.07	0.07
		10000	0.09	0.17	0.29	0.11	0.48	0.08	0.48	0.08	0.58	0.40	0.29	0.45	N/A	0.13	0.13	0.13
		12000	0.12	0.23	0.39	0.14	0.66	0.11	0.66	0.11	0.82	0.55	0.39	0.62	N/A	0.20	0.20	0.20

- (a) Tested with: 2-in filters 6 to 25 Tons.
 (b) Difference in pressure drop should be considered when utilizing optional 2-in pleated filters.
 (c) OA = Outside Air and RA = Return Air.
 (d) 6-12.5 Ton Low Leak static pressure does not differentiate between Downflow and Horizontal.

Heating Performance

Table 47. Auxiliary electric heat capacity

Unit Model Number	Total ^(a)		No. of Stages	Stage 1		Stage 2	
	kw Input ^(b)	MBh Output		kw Input	MBh Output	kw Input	MBh Output
TH*072*3,4,W	9.00	30.735	1	9.00	30.735	-	-
TH*090*3,4,W	18.00	61.47	1	18.00	61.47	-	-
TH*102*3,4,W	27.00	92.205	2	18.00	61.47	9.00	30.735
	36.00	122.94	2	18.00	61.47	18.00	61.47
TH*120*3,4,W	18.00	61.47	1	18.00	61.47	-	-
TH*150*3,4,W	27.00	92.205	2	18.00	61.47	9.00	30.735
	36.00	122.94	2	18.00	61.47	18.00	61.47
	54.00	184.41	2	36.00	122.94	18.00	61.47
TH*180*3,4,W	18.00	61.47	1	18.00	61.47	-	-
	36.00	122.94	2	18.00	61.47	18.00	61.47
	54.00	184.41	2	36.00	122.94	18.00	61.47
TH*210*3,4,W	36.00	122.94	2	18.00	61.47	18.00	61.47
TH*240*3,4,W	54.00	184.41	2	36.00	122.94	18.00	61.47
TH*300*3,4,W	72.00	245.88	2	36.00	122.94	36.00	122.94

Table 48. Air temperature rise

kW	Stages	6 Tons 1800 cfm	7.5 Tons 2250 cfm	8.5 Tons 2550 cfm	10 Tons 3000 cfm	12.5 Tons 3750 cfm
		Three Phase TH*072*3,4,W	Three Phase TH*090*3,4,W	Three Phase TH*102*3,4,2	Three Phase TH*120*3,4,W	Three Phase TH*150*3,4,W
9.00	1	15.81	12.64	11.16	-	-
18.00	1	31.61	25.29	22.31	18.97	15.17
27.00	2	47.42	37.93	33.47	28.45	22.76
36.00	2	63.22	50.58	44.63	37.93	30.35
54.00	2	-	-	-	56.90	45.52
72.00	2	-	-	-	-	-
kW	Stages	15 Tons 4500 cfm	17.5 Tons 5250 cfm	20 Tons 6000 cfm	25 Tons 7500 cfm	
		Three Phase TH*180*3,4,W	Three Phase TH*210*3,4,W	Three Phase TH*240*3,4,W	Three Phase TH*300*3,4,W	
9.00	1	-	-	-	-	
18.00	1	12.64	-	-	-	
27.00	2	-	-	-	-	
36.00	2	25.28	21.67	18.96	15.17	
54.00	2	37.93	32.51	28.45	22.76	
72.00	2	-	43.35	37.93	30.34	

Note: For minimum design airflow, see airflow performance table for each unit. To calculate temp. rise at different airflow, use the following formula:

$$\text{Temp. rise across Electric Heater} = (\text{kW} \times 3414) / (1.08 \times \text{cfm}).$$



Controls

Enhanced BAS Integration and Connectivity

- Symbio™ 700 integrates seamlessly with Trane® Tracer® Synchrony and Tracer Ensemble® to deliver optimized building automation and building management features and functions.
- Easily integrate with open standard protocols to connect seamlessly to a BAS (whether that is Trane or non-Trane).
- Digit 21 must equal 1, 2, or 3 for communication support.

BACnet® Communications

Symbio™ 700 includes native BACnet communications which allows the unit to communicate directly with a Tracer or non-Trane Building Automation System via open protocol BACnet MS/TP or IP.

Modbus Communications

Symbio 700 includes native Modbus communications which allows the unit to communicate directly with a Tracer or non-Trane Building Automation System via open protocol Modbus RTU or TCP/IP.

LonTalk® Communications

The optional LonTalk® communications module allows the unit to communicate directly with a Tracer or non-Trane Building Automation System via open protocol LonTalk.

Air-Fi® Wireless Communications

The optional Air-Fi communications module allows the unit to communicate directly with a Tracer or non-Trane Building Automation System via open protocol BACnet over Zigbee wireless.

Secure Remote Connectivity with Trane Connect

The Symbio controller enables secure remote connectivity via Trane Connect to Trane Intelligent Services and remote monitoring. Trane Connect provides anywhere/anytime access to monitor and manage with secure remote access and connectivity options through a multitude of platforms. Peace of mind that the system will be operational and provide comfort to customers.

Serviceability

Symbio Service and Installation Mobile App

The Symbio Service and Installation mobile app is accessible through mobile devices (phones and tablets) via Bluetooth connectivity or via Trane Connect. The intuitive mobile app feels natural to technicians and operators. They'll quickly be able to view equipment status and alarms, perform startup tasks, change configurations, test the equipment's performance in specific modes—and much more. Free for download from App Store (Apple iOS) and Google Play (Android devices).

To download the Symbio Service and Installation Mobile App use the links below or scan the code with your mobile phone camera.

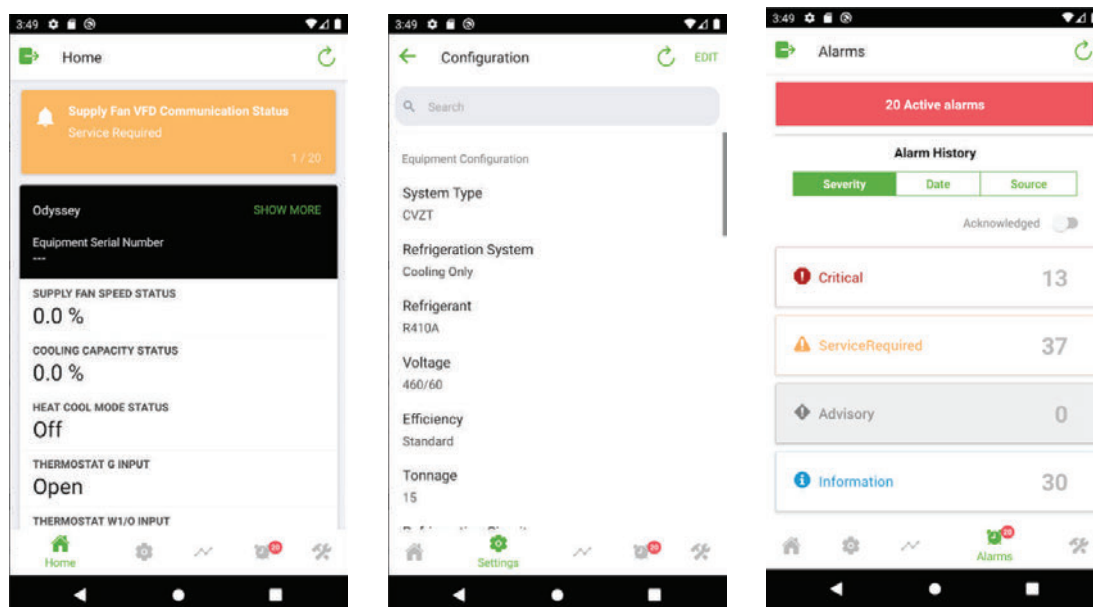
Apple download link (<https://apps.apple.com/us/app/symbio-service-installation/id1309310176>)

Google Play (Android) download link (<https://play.google.com/store/apps/details?id=com.trane.mobileservicetool>)

Figure 8. Scan code



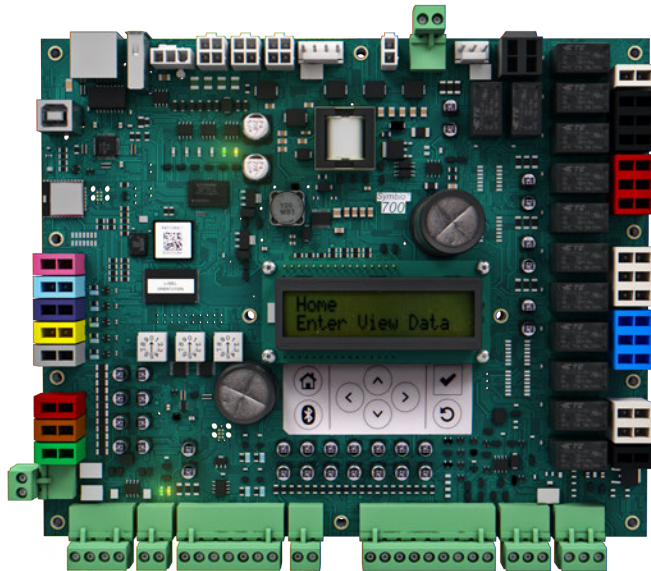
Figure 9. Symbio service and installation mobile app



Onboard User Interface

An integrated onboard user interface that makes setup and continued operation easy. It provides real time operational performance, status, data, and alarms. It also allows the user to interact with, service, troubleshoot, and control their equipment without additional service software tools or when a mobile interface is not available.

Figure 10. Onboard user interface



Service Test Mode

Symbio 700 requires no special tools to run the unit through its paces. Simply navigate to the 'Service' section of the on-board user interface or the 'Tools' section of the Symbio Service and Installation Mobile App and enter the 'Service Test Mode' section. Here the unit can be placed in the desired operating condition for a pre-determined amount of time supporting troubleshooting efforts in the field. The Symbio 700 will return to normal control when the user exits test mode or when the pre-determined, user-selected Service Test time has expired.

Symbio 700 controls with upgradeable software

Trane's equipment and systems feature engineered, tested, and proven applications that meet industry energy standards and provide the flexibility to customize and update over the life of the equipment. Professional operational algorithms are embedded within the Symbio 700 controller at the Trane factory. Symbio 700 standardizes each equipment unit to maintain standards for comfort, efficiency, and air quality, without additional field programming. Symbio 700 provides the flexibility over the life of the equipment to meet changing customer needs and/or industry standards.

Flexibility

Expansion Modules (requires Tracer® TU)

- XM30 – Provides 4 universal inputs or analog outputs
- XM32 – Provides 4 binary outputs

Field Programming via TGP2 (requires Tracer TU)

- Control ancillary equipment
- Custom sequences

TGP2 and XM Limitations:

- Programs will only have access to available BACnet® points. (Ensures system reliability.)
- TGP2 programs will not have direct I/O control access for factory components. (Compressors will not be able to be directly controlled On/Off without going through factory provided protection sequences.)
- Onboard I/O will not be available to custom applied TGP2 programs. If additional I/O is required for a new control loop, a separate expansion module will be required.

- Customer applied I/O will be limited to a maximum combination of 2 XM modules. Only XM30 or XM32 modules will be supported by the Symbio 700 UC.
- Tracer TU will be required to configure XMs and to create, view, or modify TGP2 programs.

Economizer Controls

There are four options for economizer control, Dry Bulb Temperature, Comparative Enthalpy, Reference Enthalpy and Differential Dry Bulb Temperature.

Dry Bulb Temperature Control

The dry bulb system measures outdoor temperature comparing it to the economizer enable setpoint. If the outdoor temperature is below the economizer enable setpoint, the economizer will operate freely. This system is best suited for arid regions where the humidity levels of outside air would not be detrimental to building comfort and indoor air quality.

Comparative Enthalpy Control

The comparative enthalpy system measures the temperature and humidity of both return air and outside air to determine which source has lower enthalpy. This system allows true comparison of outdoor air and return air enthalpy by measurement of outdoor air and return air temperature and humidity.

Reference Enthalpy Control

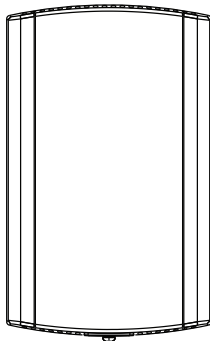
The reference enthalpy system compares outdoor air temperature and humidity to the economizer enthalpy enable setpoint. If outdoor air temperature and humidity are below the economizer enthalpy enable setpoint, the economizer will operate freely. This system provides more sophisticated control where outdoor air humidity levels may not be acceptable for building comfort and indoor air quality.

Differential Dry Bulb Temperature Control

The differential dry bulb system measures the temperature of both return air and outside air to determine when to economize. If outdoor air temperature is below the return air temperature minus a differential, the economizer will operate freely. This system is best suited for arid regions where the humidity levels of outside air would not be detrimental to building comfort and indoor air quality.

Zone Sensors

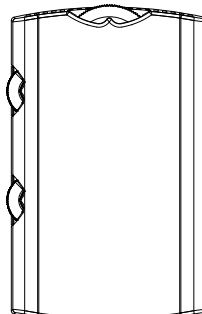
Zone Temperature Only



BAYSENS077

Provides temperature input only. Can be used as a secondary remote temperature input for thermostats.

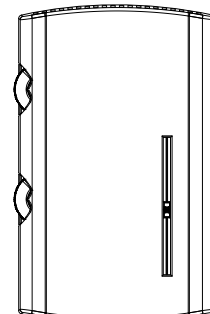
Manual Changeover



BAYSENS106

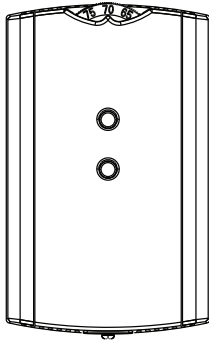
Heat, Cool or Off System Switch. Fan Auto or Off Switch. Single temperature setpoint thumbwheel.

Manual/Automatic Changeover

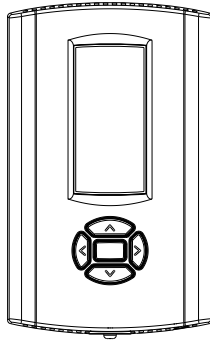


BAYSENS108

Auto, Heat, Cool or Off System Switch. Fan Auto or Off Switch. Dual temperature setpoint sliders

Integrated Comfort™ System

BAYSENS073 / BAYSENS074 / BAYSENS075

Sensor(s) available with optional temperature adjustment and override buttons to provide central control through a Trane Integrated Comfort system.

Wired Display Sensor

BAYSENS135

LCD display that provides heat, cool, auto, or off. Includes two temperature setpoints and a lockable setting with °F or °C indicators.

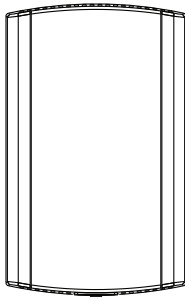
Touchscreen Digital Display Communicating Sensor

BAYSENS800

Uses BACnet® MS/TP link to communicate zone temperature and setpoints. Auto, Heat, Cool or Off System Switch. Fan Auto or On Switch. 7-day programmable thermostat with night setback.

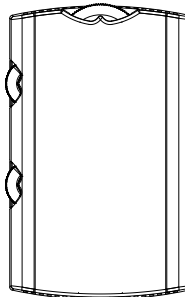
Note: Not compatible with VAV units. Requires BACnet communications.

Air-Fi Wireless Communicating Zone Sensors

Wireless Zone Temperature Only

BAYSENS077

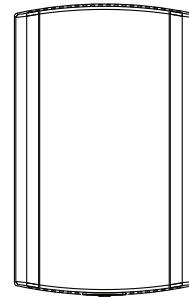
Measures temperature and optional humidity (with WCS-SH) for use in public spaces where no local user interface is preferred.

Note: Requires BACnet communications.

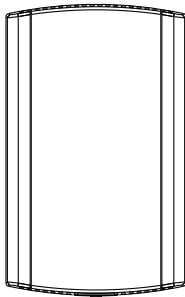
Wireless Display Sensor

BAYSENS106

Easy-to-use interface for clear and simple monitoring and control. Can be configured for any Trane system or to meet the customer's preference.

Note: Requires BACnet communications.

Wired CO₂ Sensor

X13790422010

The maintenance-free carbon dioxide (CO₂) sensor is primarily used for demand control ventilation applications.

Wired Zone Temperature and Humidity Sensor

BAYSENS036

Measures temperature and relative humidity. Relative humidity input is used to control activation of dehumidification.